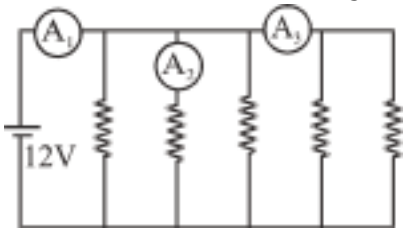
PHYSICS

1) In the circuit, each resistance is 20 Ω . The reading of A_1, A_2 and A_3 are respectively -



(all ammeters are ideal) :-

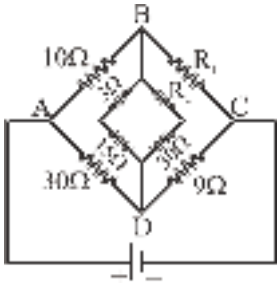
(1) 3A, 1.8A, 1.2A

(2) 3A, 1.2 A, 0.6A

(3) 3A, 0.6 A, 1.2 A

(4) 3A, 0.6 A, 0.6 A

2) In the Wheatstone bridge shown below, in order to balance the bridge, we must have :-



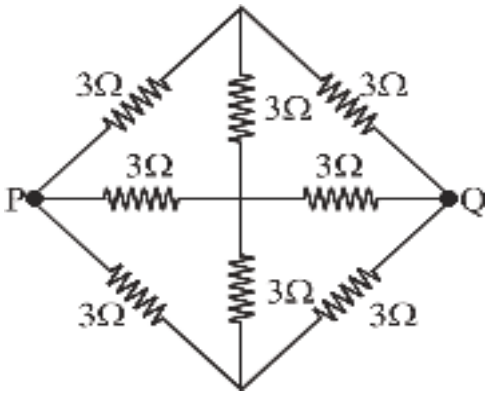
(1) $R_1 = 3\Omega$; $R_2 = 3\Omega$

(2) $R_1 = 6\Omega$; $R_2 = 15\Omega$

(3) $R_1 = 1.5\Omega$; R_2 = any finite value

(4) $R_1 = 3\Omega$; R_2 = any finite value

3) In the network shown the net resistance between points P and Q is :-

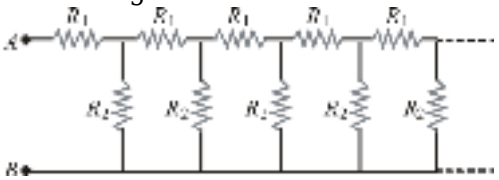


(1) 1 Ω

(2) 2 Ω

(3) 3 Ω

(4) 6 Ω

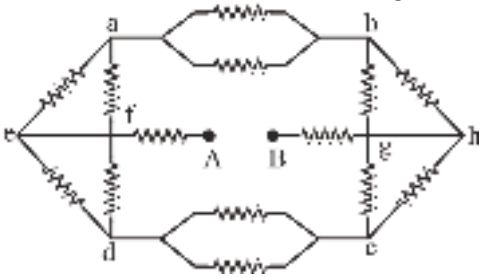
4) An infinite sequence of resistance is shown in the figure. The resultant resistance between A and B will be, when R_1 = 1 ohm and R_2 = 2 ohm 

(1) Infinity

(2) 1 Ω

(3) 2 Ω

(4) 1.5 Ω

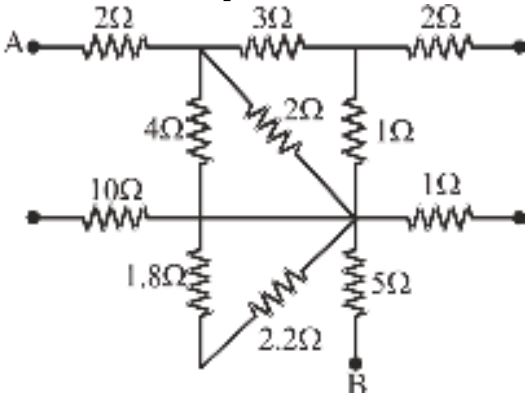
5) Each of the resistors shown in the figure has resistance R. Find the equivalent resistance between A and B : 

(1) $\frac{7R}{4}$

(2) $\frac{5R}{4}$

(3) $\frac{9R}{4}$

(4) $\frac{11R}{4}$

6) What is the equivalent resistance between the points A and B of the network : 

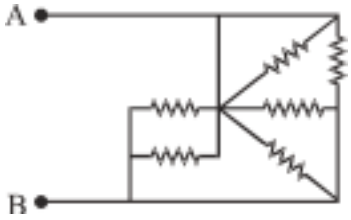
(1) $\frac{57}{7}\Omega$

(2) 8 Ω

(3) 6 Ω

(4) $\frac{57}{5}\Omega$

7) The circuit shown has six resistors of equal resistance R. Find the equivalent resistance between A and B :



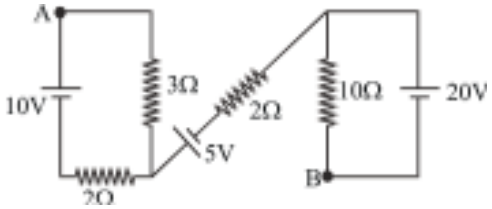
(1) $\frac{11R}{12}$

(2) $\frac{13R}{12}$

(3) $\frac{R}{5}$

(4) $\frac{15R}{12}$

8) Find potential difference between point A & B:-



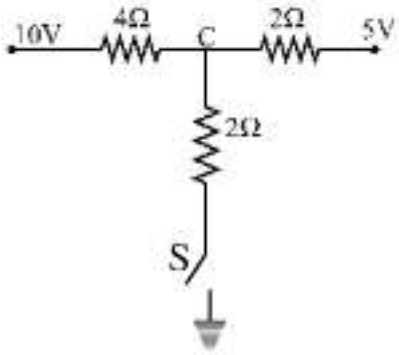
(1) 26 V

(2) 31 V

(3) 30 V

(4) 15 V

9) As the switch S is closed in the circuit shown in figure, current passed through it is



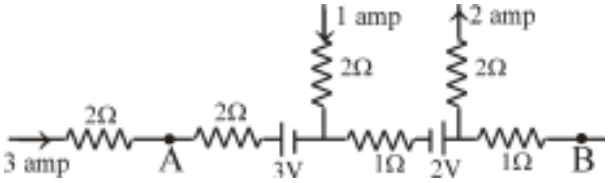
(1) zero

(2) 1 A

(3) 2 A

(4) 1.6 A

10) The potential difference between points A and B in a section of a circuit shown is :-



(1) 5 volts

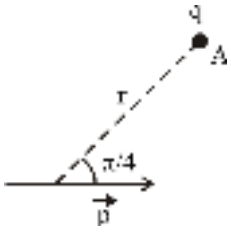
(2) –13 volt

(3) 0 volts

(4) +13 volts

11)

Dipole moment of a short dipole is p. A point charge q is placed at point A. What will be force on q at point A. :-



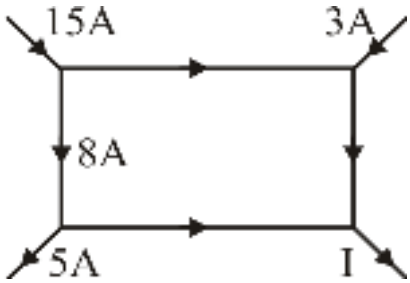
(1) $\sqrt{\frac{5}{2}}\frac{Kqp}{r^3}$

(2) $\frac{2Kpq}{r^3}$

(3) $\frac{Kpq}{r^3}$

(4) $\frac{\sqrt{3}Kpq}{2r^3}$

12) The value of current I in the given circuit is :-

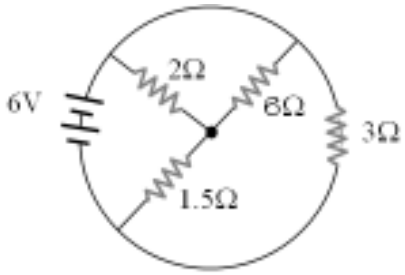


(1) 3A

(2) 13A

(3) 23A

(4) –3A



13) The total current supplied to the circuit by the battery is

(1) 1 A

(2) 2 A

(3) 4 A

(4) 6 A

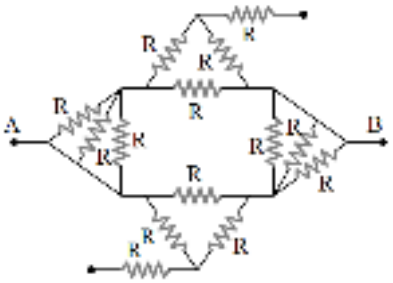
14) The specific resistance of a wire is ρ, its volume is 3 m^3 and its resistance is 3 ohms, then its length will be :

(1) $\sqrt{\frac{1}{\rho}}$

(2) $\frac{3}{\sqrt{\rho}}$

(3) $\frac{1}{\rho}\sqrt{3}$

(4) $\rho\sqrt{\frac{1}{3}}$



15) Find equivalent resistance between A and B

(1) R

(2) $\frac{3R}{4}$

(3) $\frac{R}{2}$

(4) 2R

16) A charged particle of mass m and charge q is released from rest in a uniform electric field E. Neglecting the effect of gravity, the kinetic energy of the charged particle after 't' second is

(1) $\frac{Eq^2m}{2t^2}$

(2) $\frac{2E^2t^2}{mq}$

(3) $\frac{E^2q^2t^2}{2m}$

(4) $\frac{Eqm}{t}$

17) Two opposite and equal charges 4×10^{-8} coulomb when placed 2×10^{-2} cm away, form a dipole. If this dipole is placed in an external electric field 4×10^{8} newton/coulomb, the value of maximum torque and the work done in rotating it through 180° will be

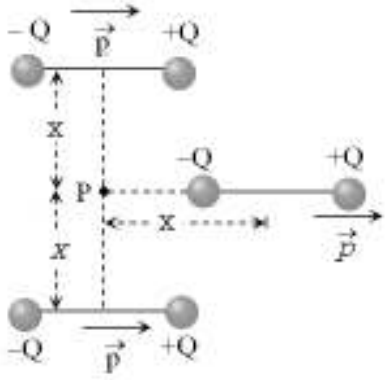
(1) 64×10^{-4}Nm and 64×10^{-4}J

(2) 32×10^{-4} Nm and 32×10^{-4}J

(3) 64×10^{-4}Nm and 32×10^{-4}J

(4) 32×10^{-4} Nm and 64×10^{-4}J

18) Three identical dipoles are arranged as shown below. What will be the net electric field at P



$\left(k = \frac{1}{4\pi\varepsilon_0}\right)$

(1) $\frac{k.p}{x^3}$

(2) $\frac{2kp}{x^3}$

(3) Zero

(4) $\frac{\sqrt{2}\,kp}{x^3}$

19) Match the column :

	Column-I		**Column-II**
(A)	$\int_0^2 5\,dx$	(P)	1
(B)	$\int_1^2 x\,dx$	(Q)	10
(C)	$\int_1^2 x^{-1}\,dx$	(R)	1.5

(D)	$\int_0^{\infty} e^{-x}\,dx$	(S)	$\log_e 2$

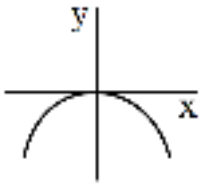
(1) A-P ; B-Q ; C-R ; D-S

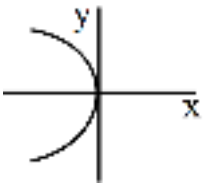
(2) A-P ; B-R ; C-Q ; D-S

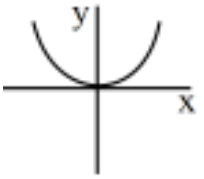
(3) A-Q ; B-R ; C-S ; D-P

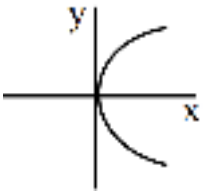
(4) A-Q ; B-R ; C-P ; D-S

20) If $y = x^2$ then y vs x curve will be -

(1) 

(2) 

(3) 

(4) 

21) $y = \frac{1+x}{e^x}$ then $\frac{dy}{dx}$ is equal to -

(1) $\frac{x}{e^x}$

(2) $-\frac{x}{e^x}$

(3) $\frac{(x+1)}{e^x}$

(4) None of these

22) Value of $\int_3^5 \frac{1}{2x+3}\,dx$ is :

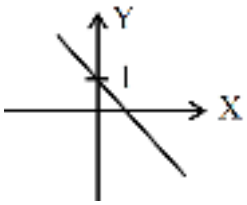
(1) $\ell n \frac{13}{9}$

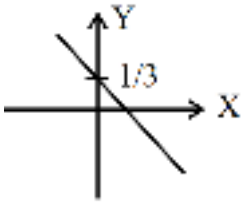
(2) $\frac{1}{2}\ell n\frac{13}{9}$

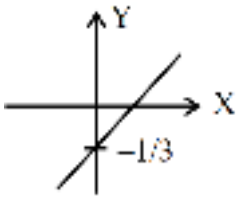
(3) $\frac{1}{2}\ell n\frac{15}{9}$

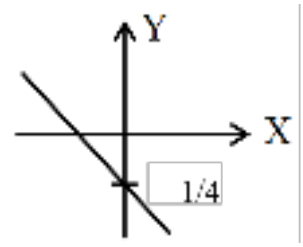
(4) None of these

23) Correct graph of 3x + 4y + 1 = 0 is -

(1) 

(2) 

(3) 

(4) 

24) At what angle should the two forces 2P and $\sqrt{2}P$ act so that the resultant force $P\sqrt{10}$ is

(1) 45°

(2) 60°

(3) 90°

(4) 120°

25)

Two vector $\vec{A}$ and $\vec{B}$ are such that $|\vec{A}+\vec{B}|=|\vec{A}-\vec{B}|$. The angle between $\vec{A}$ and $\vec{B}$ will be:

(1) 0

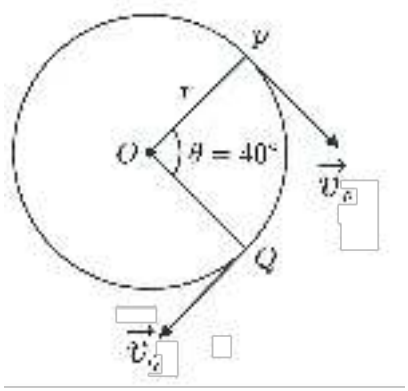
(2) π/3

(3) π/2

(4) π

26)

A particle is moving on a circular path of radius r with uniform velocity v. The change in velocity when the particle moves from *P* to *Q* is $(\angle POQ = 40^o)$



(1) $2v\cos 40^o$

(2) $2v\sin 40^o$

(3) $2v\sin 20^o$

(4) $2v\cos 20^o$

27) The angle between $\vec{A}$ and $\vec{B}$ is θ. The value of the triple product $\vec{A}.(\vec{B}\times\vec{A})$ is

(1) A^2B

(2) Zero

(3) $A^2B \sin\theta$

(4) $A^2B \cos\theta$

28) The angle between two vectors $-2\hat{i}+3\hat{j}+\hat{k}$ and $\hat{i}+2\hat{j}-4\hat{k}$ is :

(1) 0º

(2) 90º

(3) 180º

(4) None of the above

29) A person blows into open-end of a long pipe. As a result, a high-pressure pulse of air travels down the pipe. When this pulse reaches the other end of the pipe, then correct statements are :-
(A) a high-pressure pulse starts travelling up the pipe, if the other end of the pipe is open
(B) a low -pressure pulse starts travelling up the pipe, if the other end of the pipe is open
(C) a low pressure pulse starts travelling up the pipe, if the other end of the pipe is closed
(D) a high-pressure pulse starts travelling up the pipe, if the other end of the pipe is closed

(1) A,B

(2) B,C

(3) B,D

(4) A, D

30) A tuning fork vibrating with a sonometer having 20 cm wire produces 5 beats per sec. The beat frequency does not change if the length of the wire is changed to 21 cm. The frequency of the tuning fork must be :-

(1) 200 Hz

(2) 210 Hz

(3) 205 Hz

(4) 215 Hz

31) An organ pipe P closed at one end vibrates in its first harmonic. Another organ pipe Q open at both ends vibrates in its third harmonic. When both are in resonance with a tuning fork, the ratio of the length of P to that of Q is :-

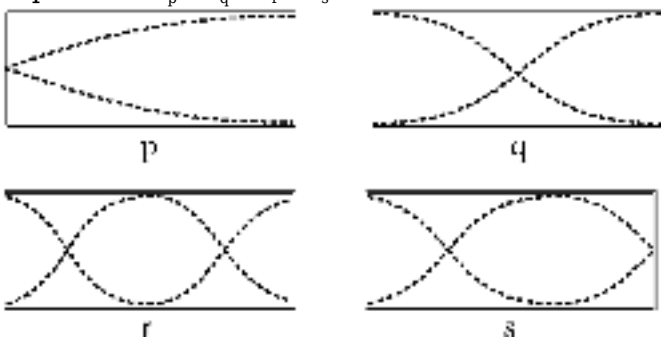
(1) $\frac{1}{2}$

(2) $\frac{1}{4}$

(3) $\frac{1}{6}$

(4) $\frac{1}{8}$

32) The vibrations of four air columns of same length are represented in figure. The ratio of frequencies $n_p : n_q : n_r : n_s$ is:



(1) 12 : 6 : 3 : 4

(2) 1 : 2 : 4 : 3

(3) 4 : 2 : 3 : 1

(4) 6 : 2 : 3 : 4

33) A student is experimenting with resonance tube apparatus in physics lab to find the speed of sound at room temperature. He got 1st two resonating lengths of air column as 17 cm and 51 cm, using tuning fork of frequency 512 Hz. Find speed of sound at room temperature :-

(1) 348 m/s

(2) 358 m/s

(3) 332 m/s

(4) 322 m/s

34) The displacement of the particle is given by $y = a + bt + ct^2 - dt^4$. The initial velocity and initial acceleration are respectively

(1) b, –4d

(2) –b, 2c

(3) b, 2c

(4) 2c, –4d

35) A particle is moving eastwards with velocity of 5 m/s. In 10 sec the velocity changes to 5 m/s northwards. The average acceleration in this time is

(1) Zero

(2) $\frac{1}{\sqrt{2}}\ m/s^2$ toward north-west

(3) $\frac{1}{\sqrt{2}}\ m/s^2$ toward north-east

(4) $\frac{1}{2}\ m/s^2$ toward north-east

36) A particle moves along a straight line such that its displacement at any time t is given by :
$s = \left(t^3 - 3t^2 + 2\right)\ m$
The displacement when the acceleration becomes zero is:

(1) 0 m

(2) 2 m

(3) 3 m

(4) –2 m

37) A body A starts from rest with an acceleration a_1. After 2 seconds, another body B starts from rest with an acceleration a_2. If they travel equal distances in the 5th second, after the start of A, then the ratio $a_1 : a_2$ is equal to

(1) 5 : 9

(2) 5 : 7

(3) 9 : 5

(4) 9 : 7

38) The ratio of displacement in n second and in the nth second for a particle moving in a straight line under constant acceleration starting from rest is:

(1) $\frac{2n-1}{n^2}$

(2) $\frac{1}{n}$

(3) $\frac{n^2}{n-1}$

(4) $\frac{n^2}{2n-1}$

39) **Assertion:** A body is momentarily at rest at the instant, if it reverse the direction.

Reason: A body cannot have acceleration, if its velocity is zero at a given instant of time.

(1) Assertion and Reason both are correct and Reason is the correct explanation of Assertion.

(2) Assertion and Reason both are correct but Reason is not the correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

40) A car moving with a speed of 40 km/h can be stopped by brakes after atleast 4m. If it has a speed of 100 km/h, the minimum stopping distance is

(1) 16 m

(2) 20 m

(3) 25 m

(4) 35 m

41) A particle is moving on a circular path of radius 2m. Find out its displacement when it covers angle $\frac{2\pi}{3}$ radian.

(1) $2\sqrt{3}$m

(2) $\frac{2}{\sqrt{3}}$m

(3) 2m

(4) $\sqrt{3}$m

42) Let $\vec{v}$ and $\vec{a}$ denote the velocity and acceleration respectively of a body in one-dimensional motion

(1) $|\vec{v}|$ must decrease when $\vec{a} < 0$

(2) speed must increase when $\vec{a} > 0$

(3) speed will increase when both $\vec{v} < 0$ and $\vec{a} > 0$

(4) speed will decrease when $\vec{v} < 0$ and $\vec{a} > 0$

43)

A body thrown vertically upwards with an initial velocity u reaches maximum height in 6 seconds. The ratio of the distance travelled by the body in the first second and the seventh second is :- (g = 10 m/s^2)

(1) 1 : 1

(2) 11 : 1

(3) 1 : 2

(4) 1 : 11

44) A ball released from large height falls freely towards earth. Another body is released from the same height exactly one second later. The separation between the two bodies after two second the release of the second body is :-

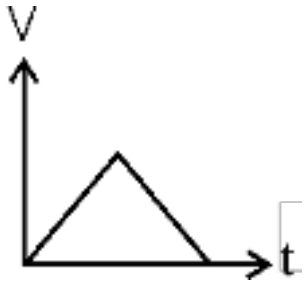
(1) 15 m

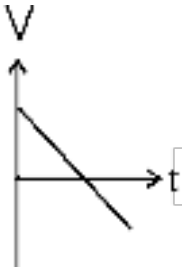
(2) 35 m

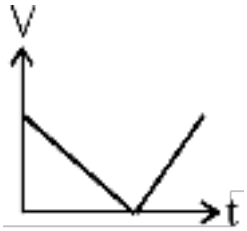
(3) 25 m

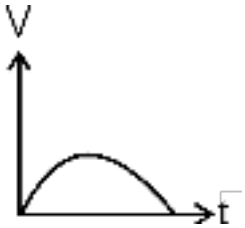
(4) 20 m

45) A body is projected vertically upward from the ground, then the velocity time graph is :

(1) 

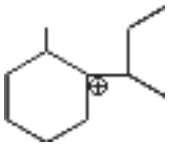
(2) 

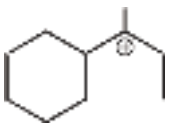
(3) 

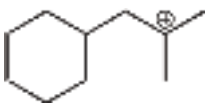
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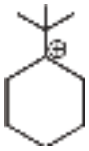
CHEMISTRY

1) Which carbocation is most stable ?

(1) 

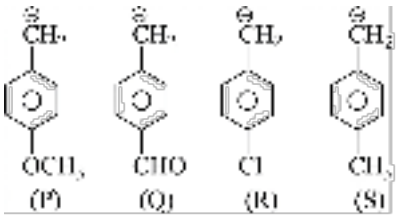
(2) 

(3) 

(4) 

2)

The decreasing order of stability of following anions is :-



(1) Q > R > S > P

(2) R > Q > P > S

(3) S > P > R > Q

(4) P > Q > R > S

3) Which can not be reduced by Zn + HCl

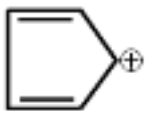
(1) CH_3CH_2 – F

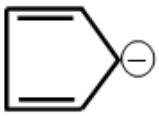
(2) CH_3CH_2 – Cl

(3) CH_3CH_2 – Br

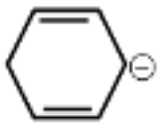
(4) CH_3CH_2 – I

4) Which shows aromatic character :

(1) 

(2) 

(3) 

(4) 

5) **Statement-I :** Phenol is million time more acidic than ethanol.
Statement-II : CF_3COOH is stronger acid than CH_3COOH.

(1) Statement-I is correct but statement-II is incorrect

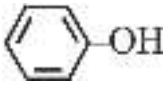
(2) Statement-I is incorrect but statement-II is correct

(3) Both statement-I and statement-II are correct

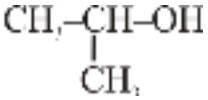
(4) Both statement-I and statement-II are incorrect

6) Which has least bond length in C–O bond ?

(1) CH_3–OH

(2) 

(3) C_2H_5–OH

(4) 

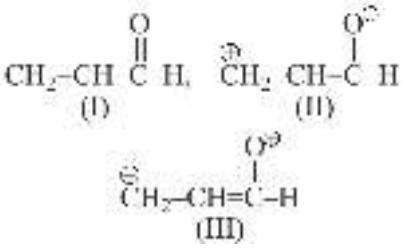
7) **Statement-I :** $PhCH_2NH_2$ is more basic than NH_3.
Statement-II : $(C_2H_5)_2NH$ is more basic than $(CH_3)_2NH$.

(1) Statement-I is correct but statement-II is incorrect

(2) Statement-I is incorrect but statement-II is correct

(3) Both statement-I and statement-II are correct

(4) Both statement-I and statement-II are incorrect

8) 

Correct order of stability of resonating structure is :-

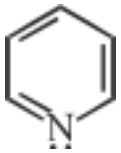
(1) I > II > III

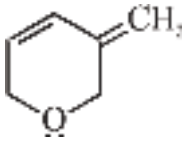
(2) III > I > II

(3) III > II > I

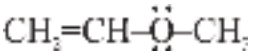
(4) I = II = III

9) In which of the following lone-pair is involved in resonance :-

(I) 

(II) 

(III) 

(IV) 

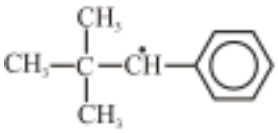
(1) I, III

(2) II, IV

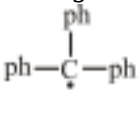
(3) III, IV

(4) I, II

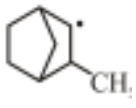
10) In which of the following radical hyper conjugation not occurs :-



(A)



(B)



(C)

(1) A & B

(2) A & C

(3) B & C

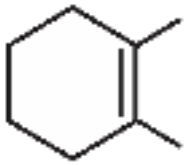
(4) A, B & C

11) Arrange the following in the stability order

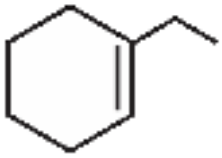
(I)



(II)



(III)



(1) I < II < III

(2) II < I < III

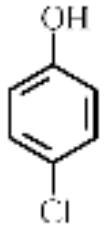
(3) I < III < II

(4) II < III < I

12)

Arrange following compounds in order of decreasing acidity :-

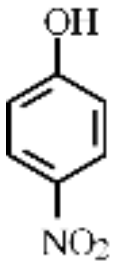
(I)



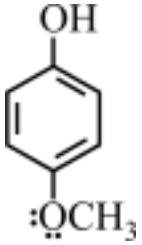
(II)



(III)



(IV)



(1) I > II > III > IV

(2) III > I > II > IV

(3) III > II > I > IV

(4) I > III > II > IV

13) The resonance structure have

(1) The same positions of nuclei

(2) The same number of unpaired electron

(3) Both (1) and (2)

(4) Neither same position of nuclei nor the same number of unpaired electron

14)

Which is incorrect order of basic strength ?

(1) $CH_3–CH_2CH_2–NH_2 > CH_3–CH_2–NH_2 > CH_3–NH_2$

(2)  $NH_2 >$  $CH_2–NH_2 > CH_3–CH_2–NH_2$

(3) $CH_3–CH_2–NH_2 > CH_3–CH{=}NH: > CH_3–C{\equiv}N$

(4) $CH_3–NH–CH_3 > CH_3–CH_2–NH_2 > CH_3–NH_2$

15) The transfer of electrons away from an atom or substituent group attached to the conjugated system is known as -

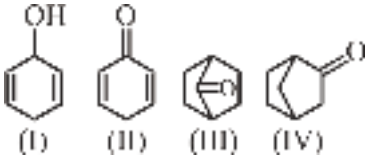
(1) Temparary effect

(2) Positive resonance effect

(3) Negative resonance effect

(4) None of the above

16) Tautomerism is exhibited by :-

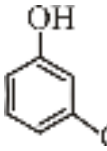
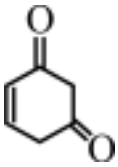


(1) II, III, IV

(2) I, II, IV

(3) II, IV

(4) III, IV

17) **Assertion :**  Enol is more stable than  Keto.
Reason : Enol is stabilised by aromaticity.

(1) Assertion and Reason are true but not correct explanation.

(2) Assertion is true but Reason is false.

(3) Assertion is false but Reason is true.

(4) Assertion and reason are true and correct explanation.

18) 1 Mole of liquid A and 2 moles of liquid B make a solution having observed vapour pressure of 42 torr. The vapour pressure of pure A and pure B are 45 torr and 36 torr respectively. The described

solution :-

(1) is an ideal solution

(2) shows negative deviation

(3) is a minimum boiling azeotrope

(4) has volume less than the sum of individual volumes of both components

19) A solution weighing 'a' gm has molality b. Find the molecular mass of solute if the mass of solute is 'c' gm :-

(1) $\frac{c}{b} \times \frac{1000}{(a-c)}$

(2) $\frac{b}{a} \times \frac{1000}{(a-b)}$

(3) $\frac{b}{c} \times \frac{1000}{(a-c)}$

(4) $\frac{c}{a} \times \frac{1000}{(b-a)}$

20) The value of K_b is given by $RT_b^2/1000$ X. In this relation X is :-

(1) Density of solution

(2) $\Delta H°f$ of solvent

(3) Latent heat of vaporisation

(4) Enthalpy of vaporisation

21) A solution has 1 : 4 mole ratio of pentane to hexane. The vapour pressures of the pure hydrocarbons at 20°C are 440 mm Hg for pentane and 120 mm Hg for hexane. The mole fraction of pentane in the vapour phase would be :-

(1) 0.200

(2) 0.478

(3) 0.549

(4) 0.789

22) K_f for water is 1.86 K kg mol^{-1}. If automobile radiator holds 1.0 kg of water, how many grams of ethylene glycol ($C_2H_6O_2$) must be added to get the freezing point of the solution lowered to –2.8°C ?

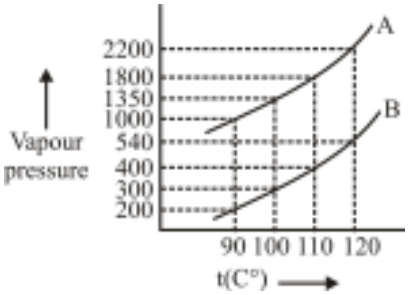
(1) 27g

(2) 72 g

(3) 93 g

(4) 39 g

23) Determine the boiling point of a mixture containing 1000 gm A (molar mass = 50) and 700 g B (molar mass = 70) using the following graph at atmospheric pressure of 1000 torr :-



(1) 90°

(2) 100°C

(3) 110°C

(4) 120°C

24) 10 ml of A (boiling point = 68°C) and 10 ml of B (B.pt = 95°C) were mixed together and 19.95 ml of mixture was obtained. The B.pt of azeotropic mixture obtained by them will be :

(1) < 68°C

(2) > 68°C but < 95°C

(3) > 95°C

(4) Unpredictable

25) An ideal solution is formed by mixing two volatile liquids A and B. X_A and X_B are the mole fractions of A and B respectively in the solution and Y_A and Y_B are the mole fractions of A and B respectively in the vapour phase. A plot of $\frac{1}{Y_A}$ along y-axis against $\frac{1}{X_A}$ along x-axis gives a straight line. What is the slope of the straight line ?

(1) $\frac{P_B^\circ}{P_A^\circ}$

(2) $\frac{P_A^\circ}{P_B^\circ}$

(3) $P_B^\circ - P_A^\circ$

(4) $P_A^\circ - P_B^\circ$

26) **Assertion :** $\Delta H_{mix.}$ & $\Delta V_{mix.}$ for the preparation of ideal solution is zero.
Reason : A-B interaction in ideal solution are same as between A-A & B-B in the two liquids before mixing.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

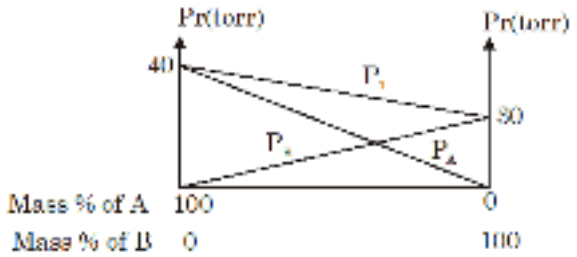
(4) Both Assertion and Reason are false.

27)

Equal mass of A & B are present in liquid solution then total pressure exerted by vapours is X torr. $\frac{X}{6}$

is :

Molar mass of A = 80gm / mol and Molar mass of B = 120gm / mol



(1) 36

(2) 5

(3) 6

(4) 7

28) The henry constant for N_2 and O_2 gas dissolved in water at 298K are 6×10^9 N/m^2 and 2.5×10^9 N/m^2. A sample of water at 298K was equilibrium with air containing 20 mole % oxygen and 80 mole % nitrogen at 1 atm. If the dissolved gas was separated from a sample of this water and then dried. Then calculate the mole % of oxygen gas in the dried sample :

(1) 37.5%

(2) 8×10^{-6}

(3) 8×10^{-4}

(4) 67%

29)

	List-I		List-II
(P)	Propan-2-ol + 2-methyl propanol	(1)	Maximum boiling azeotrope
(Q)	$H_2O + HNO_3$	(2)	Raoult's law is obeyed
(R)	$CS_2 + CH_3COCH_3$	(3)	Positive deviation from Raoult's law
(S)	$H_2O + HCOOH$	(4)	Minimum boiling azeotrope
		(5)	Mixture form ideal solution.

(1) P-1 ; Q-3 ; R-2 : S-4

(2) P-2 ; Q-1 ; R-3 : S-1

(3) P-3 ; Q-1 ; R-3 : S-2

(4) P-2 ; Q-4 ; R-1 : S-3

30) In which of the following pairs of solutions will the values of the vant Hoff factor be the same?
(1) 0.05 M K_4 $[Fe(CN)_6]$ and 0.10 M $FeSO_4$
(2) 0.10 M $K_4[Fe(CN)_6]$ and 0.05 M $FeSO_4$ $(NH_4)_2SO_4$. $6H_2O$
(3) 0.20 M NaCl and 0.10 M $BaCl_2$

(4) 0.05 M $FeSO_4$ $(NH_4)_2SO_4$. $6H_2O$ and 0.02 M KCl . $MgCl_2$. $6H_2O$

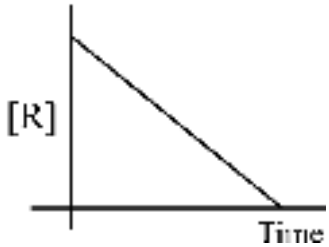
(1) 1, 2, 3, 4
(2) 2, 4
(3) 2 only
(4) 4 only

31) For the reaction $A_{(aq)} \rightarrow 2B_{(aq)} + 3C_{(aq)}$, K = 10 M/Min. If initial concentration of A is 20M then after how many seconds concentration of C becomes 45M.

(1) 1.5
(2) 2.5
(3) 60
(4) 90

32) If $T_{x\%}$ denotes time required to complete x% of a first order reaction and
$T_{75\%} = A \times T_{50\%}$, $T_{87.5\%} = B \times T_{50\%}$, $T_{99\%} = C \times T_{90\%}$
then calculate value of A + B + C.

(1) 7
(2) 6
(3) 5
(4) 4

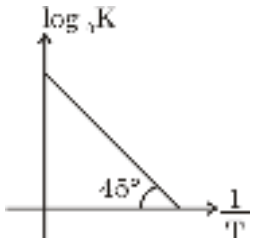
33) 

Above graph is for a reaction which have only single reactant (R) and order of reactions depends only on that single reactant then possible order of reaction may be -

(1) First order reaction
(2) 1/2 order reaction
(3) – 2 order reaction
(4) zero order reaction

34) A reaction is carried out at 600K, the same reaction when carried out in the presence of a catalyst, had the same rate at 500 K. The activation energy of reaction, assuming the catalyst lowers the energy barrier by 20 kJ/mol is -

(1) 100 kJ /mol
(2) 120 kJ /mol
(3) 80 kJ /mol
(4) 120 kcal/mol

35) 

For a reaction $\log_{10} K$ & $\frac{1}{T}$ curve is given as then value of activation energy in cal/mol will be

(1) 2
(2) 4.606
(3) 7
(4) 4

36) For a first order reaction, rate constant is given by $\log k = 14 - \frac{1.2 \times 10^4}{T}$, then what will be the value of temperature if the rate constant is 100 min^{-1} ?

(1) 100 K
(2) 327 K
(3) 1000 K
(4) 720 K

37) For a given reaction, rate = $k[A][B]^{2/3}$. Correct option(s) is/are :-
(1) Units of k = $mol^{-5/3}$ $L^{5/3}$ sec^{-1}
(2) If $[B]_0$ is large compare to $[A]_0$ then reaction obey first order kinetics
(3) On diluting the solution 8 times rate will become 32 times the initial rate
(4) The reaction is a complex reaction

(1) 1, 2
(2) 1, 4
(3) 2, 4
(4) 2, 3

38) In first order reaction (A → P) calculate $\frac{t}{t_{1/2}}$ if $\frac{15}{16}$ fraction of reactant is decayed in 't' time where $t_{1/2}$ is half life.

(1) 2
(2) 4
(3) 1
(4) 8

39) The energy of activation for a reaction is 10 kJ/mol. The presence of catalyst lowers the energy of activation by 75%. Factor by which rate of reaction increases at 27°C due to catalyst is e^x, then x is : (Take R = 25/3 J/mol-K)

(1) 3

(2) 2

(3) 5

(4) 1

40) For a first order reaction choose the **CORRECT** statement :–

(1) The degree of dissociation is equal to $\left(1-e^{-Kt}\right)$

(2) The pre-exponential factor in the arrhenius equation has the dimension of time^{-1}

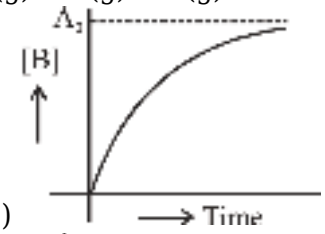
(3) A plot of reciprocal concentration of the reactant v/s time gives a straight line

(4) (1) & (2) both

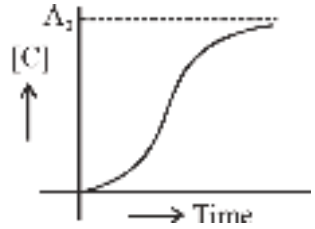
41) Which is/are correct graphs for following first order reaction :

A(g) → B(g) + C(g)

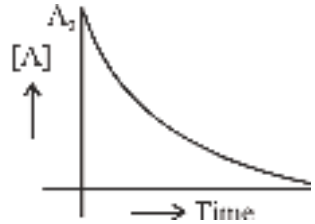
(1)



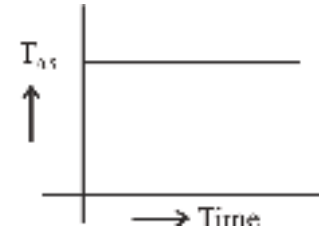
(2)



(3)



(4)



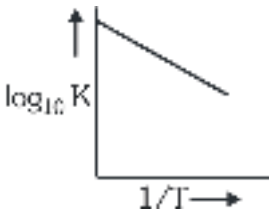
(1) 1, 3, 4 only

(2) 4 only

(3) 3, 4 only

(4) 2, 4 only

42) A graph is plotted between $\log_{10}K$ vs $\frac{1}{T}$ (K = rate constant and T → temperature). If the line makes an angle of $\tan^{-1}\left(-\frac{1}{2.303}\right)$ with (+)ve x-axis then find E_a for the reaction (in calorie) :



(1) 2

(2) 4.606

(3) 2.303

(4) 18.414

43) Given below are two statements: **Statement I :** Only those collisions in which molecules have correct orientation and sufficient energy lead to compound formation.
Statement II : Threshold energy = Activation energy + energy possessed by reacting species.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct and Statement II is incorrect.

(3) Statement I is incorrect and Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

44) The Henry's law constant for solubility of N_2 gas in water at 298 K is 1×10^5 atm. The mole fraction of N_2 in air is 0.8. Then number of moles of N_2 of air dissolved in 10 moles of water at 298 K and 5 atm is :-

(1) 4×10^{-4}

(2) 4.5×10^{-2}

(3) 4×10^{-5}

(4) 2.7×10^{-2}

45) P_A = (235 y – 125 xy) mm of Hg. P_A is partial pressure of A, x is mole fraction of B in liquid phase in the mixture of two liquids A and B and y is the mole fraction of A in vapour phase, then $P°_B$ in mm of Hg is :

(1) 235

(2) 0

(3) 110

(4) 125

BIOLOGY

1) In prokaryotes, the size of ribosome is about -

(1) 15-20 mm

(2) 15-20 nm

(3) 15-20 µm

(4) 15-20 pm

2) Which of the following statement is **incorrect** w.r.t. cilia and flagella?

(1) Cilia and flagella are membrane bound extensions of the plasma membrane

(2) The arrangement of microtubules is referred to as 9+0 organisation.

(3) Cilium and flagellum, both arise from the basal body

(4) They are cylindrical structures and their core is called an axoneme.

3) Read the following statements :-
(A) Both the chromatids of chromosome are joined by a structure called centromere
(B) During cell division spindle fibres are not formed
(C) Telocentric chromosome has a terminal centromere
(D) Every chromosome essentially has a secondary constriction
Find out correct sequence of true (T) and false (F) statements :-

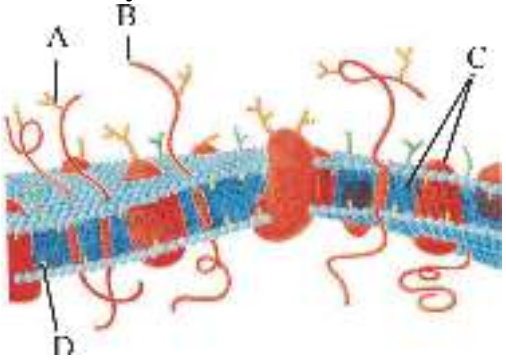
(1) T F T T

(2) F T F T

(3) T F T F

(4) T T F F

4) Identify the A, B, C and D select the correct statements regarding to A, B, C and D :-



(1) Percentage of "A" is maximum in the membrane

(2) Percentage of "B" is minimum in the membrane

(3) Lateral movement of "A" help in the measurement of fluidity of membrane

(4) The fluidity of membrane is due to "C"

5) The names of different cell organelles / structures are given below -
Mitochondria, Cell wall, Ribosomes, Centrioles, Microbodies, Plastids, Endoplasmic reticulum
How many of the above are present in both higher plant and animal cells?

(1) 3

(2) 4

(3) 5

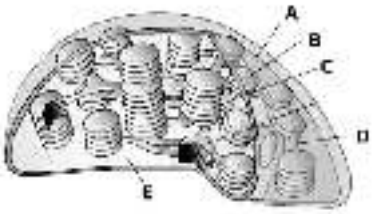
(4) 6

6) Which of the following statement are correct ?
(A) The number of chloroplast varies from per cell of *Chlamydomonas*, a green alga to 20-40 per cell

in the mesophyll.
(B) Out of the two membranes of chloroplast and mitochondria, the inner membrane is relatively less permeable.
(C) Thylakoids are arranged in stacks like the piles of coins called grana (singular : granum)
(D) The stroma of the chloroplast contains enzymes required for the synthesis of carbohydrates and proteins.

(1) A, B & C only
(2) B, C & D only
(3) A, C & D only
(4) A, B, C & D

7) Consider the following five statements (A to E) w.r.t. choroplast shown below. Select the correct



option stating which ones are True (T) and which ones are False (F):
A. It is impermeable and lacks porins.
B. It is selectively permeable, having carrier proteins for transport.
C. Stacked thylakoids one over other which is the site of production of assimilatory power.
D. Present between two grana and contains enzymes of dark reaction.
E. It contains enzymes for the synthesis of sugar and proteins.

(1) A-F B-T C-T D-T E-T
(2) A-F B-T C-T D-F E-T
(3) A-T B-F C-T D-T E-T
(4) A-T B-F C-F D-T E-T

8) Arrange the following events of meiosis in the correct sequence :-
(A) Terminilisation of chiasmata
(B) Crossing over
(C) Synapsis
(D) Disjunction of homologous chromosomes
(E) Dissolution of synaptonemal complex
The correct sequence is :-

(1) A → B → C → D → E
(2) E → D → C → B → A
(3) C → B → D → E → A
(4) C → B → E → A → D

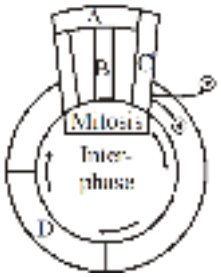
9) Which of the following is correct sequence of amount of DNA in different stages of meiosis if amount of DNA in G_1 phase is 64 pg ?

(1) Anaphase-I = 128 pg, Metaphase-I = 128 pg, Metaphase-II = 64 pg
(2) G_2 = 128 pg, Diakinesis = 256 pg, Metaphase-I = 128 pg

(3) Pachytene = 256 pg, Metaphase-I = 256 pg, Diplotene = 256 pg

(4) S = 128 pg, Diakinesis = 256 pg, Metaphase-I = 128 pg

10) The given figure is a schematic break-up of the phases/stages of cell cycle. Select the correct option regarding it :



(1) "A" represents karyokinesis which is a division of cytoplasm.

(2) "B" is telophase which is just reverse of prophase

(3) "C" is the best phase to count total number of chromosomes in any species

(4) In "D" stage, replication of DNA takes place on the template of the existing DNA

11) What is not true about cell cycle:
a. During G_1 phase there is active synthesis of RNA and proteins but no change in its DNA content.
b. In synthesis or S phase, each chromosome carries a duplicate set of genes
c. During G_2 phase, a cell contains double the amount of DNA present in the original diploid cell
d. In S-phase, a cell doubles the original diploid (2n) chromosome number

(1) c & d

(2) b & c

(3) d only

(4) b, c & d

12) When a cell is to divide it synthesized its various components including the genetic materials during interphase. Choose the correct option for synthesis of materials during interphase ?

(1) All types of DNAs of cell (nuclear and cytoplasmic) synthesis durig S-phase only

(2) Cytoplasmic growth occurs during G_1-phase only.

(3) Protein synthesis occurs during the all sub phases of interphase

(4) All the events of interphase are not genetically controlled

13)

Choose the incorrect statement :-
(A) Cells in G_0 stage are metabolically inactive.
(B) Meiosis involves two cycle of DNA replication
(C) Mitosis restores the nucleo–cytoplasmic ratio of cells
(D) Duration of cell cycle may vary from organism to organism and also from cell to cell.

Options :-

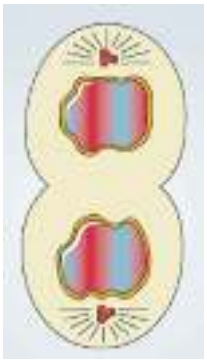
(1) A,C

(2) C,D

(3) B,C

(4) A,B

14) Identify the correct match from the column-I, II and III.

Column I		Column II		Column III	
A	Late prophase	a.		i.	ER, Golgi complex disappear
B	Anaphase	b.		ii.	Chromosome decondense and loose their individuality
C	Telophase	c.		iii.	Longest phase of meiosis-I
D	Prophase-I	d.		iv.	Splitting of centromere occur

(1) B-a-iii, A-c-iv, C-b-i, D-d-ii

(2) A-b-i, B-a-iv, C-d-ii, D-c-iii

(3) A-d-iii, B-b-iv, C-b-i, D-d-ii

(4) B-a-iii, A-c-iv, C-a-ii, D-d-i

15) Microbodies are found in–

(1) Plants only

(2) Animals only

(3) Plants and animals both

(4) Bacteria

16) Which of the following statements are true regarding the internal structure of flowering plants?
A. Study of internal structure of plant is known as plant anatomy
B. Only internal structures show adaptations for diverse environment
C. Plants have tissue as basic structural and functional unit
D. Different organs in plants show differences in their internal structure

(1) A and D
(2) A, C and D
(3) A, B, C and D
(4) Only A

17) Which of the following is common character of root hair and stem hair both?

(1) Both are multicellular
(2) Both are always hair like
(3) Both are included in epidermal tissue system
(4) Both are green and photosynthetic

18) **Assertion (A) :** Bulliform cells present in adaxial epidermis of isobilateral leaves are involved in reduction in transpiration loss of water.
Reason (R): When bulliform cells are turgid due to water stress, they make the leaves curl towards inner side.

(1) A and R both are true and R is correct explanation of A
(2) A and R both are true and R does not explain A
(3) A is true but R is false
(4) A is false but R is true

19) Ground tissue system in root and stem is differentiated in

(1) Cortex and pith only
(2) Cortex, pericycle and vascular bundle
(3) Cortex, pericycle, vascular bundle and pith
(4) Cortex, pericycle and pith

20) A transverse section of young stem of a plant is observed under the microscope. How would you ascertain that it is a dicot stem?

(1) It has scattered vascular bundles
(2) Water filled cavity is observed in it
(3) Semilunar patches of sclerenchyma is observed just above to phloem
(4) Sclerenchymatous bundle sheath is present in it

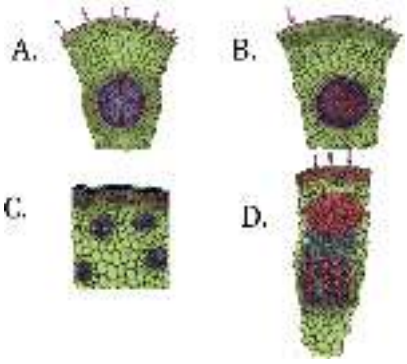
21) A dicot root can be differentiated from monocot root in having?

(1) Ring arrangement of vascular bundle
(2) Two to four xylem bundles

(3) Open vascular bundle

(4) All of the above

22) Study the following diagram carefully and identify the structure/s in which casparian strips are observed and select the correct option from options given below:



(1) A and B

(2) A and D

(3) A, B and D

(4) A, C and D

23) **Statement A :** Flowering plants exhibit enormous variation in shape, size, structure, mode of nutrition, life span, habit and habitat. **Statement B :** The flowering plant is described beginning with its habit, vegetative characters – roots, stem and leaves and then floral characters inflorescence and flower parts

(1) Statement A and B both are true

(2) Statement A is true while B is false

(3) Statement B is true while A is false

(4) Statement A and B both are false

24)

What is the position of the ovary in given diagram with respect to sepals, petals and stamens.



(1) Superior

(2) Inferior

(3) Half superior

(4) Half inferior

25) If two leaves arise from same node it is known as ….A…and it is present in…….B……

(1) A: Alternate B: *Alstonia*

(2) A: Opposite B: *Alstonia*

(3) A: Whorled B: *Calotropis*

(4) A: Opposite B: *Calotropis*

26) Match the column I with column II and select the correct option from options given below

	Column I (Plant)		**Column II (Placentation)**
A	Sunflower	G	Basal
B	Tomato	H	Axile
C	Pea	I	Marginal
D	Primrose	J	Free central

(1) A: G; B: H; C: I; D: J

(2) A: G; B: H; C: J; D: I

(3) A: G; B: I; C: H; D: J

(4) A: H; B: G; C: J; D: I

27) In how many of the following epipetalous stamens are present in flowers?
Potato, mustard, petunia, pea, tomato, bean lily, chilli, brinjal, Tobacco

(1) Five

(2) Four

(3) Six

(4) Seven

28) Which of the following is not a function of stem?

(1) Vegetative propagation

(2) Food storage

(3) Water and mineral absorption

(4) Conduction of water mineral and photosynthate

29) **Assertion (A) :** In mustard plant, flowers are bisexual and complete.
Reason (R) : In mustard, flower calyx, corolla, androecium and gynoecium are present.

(1) A and R both are true and R is correct explanation of A

(2) A and R both are true and R does not explain A

(3) A is true but R is false

(4) A is false but R is true

30) Drupe fruit in mango develops from:

(1) Monocarpellary gynoecium with inferior ovary

(2) Monocarpellary gynoecium with superior ovary

(3) Polycarpellary gynoecium with inferior ovary

(4) Polycarpellary gynoecium with superior ovary

31) Which of the following is NOT a secondary metabolite?

(1) Alkaloids

(2) Terpenoides

(3) Carbohydrates

(4) Essential oils

32) Which of the following is NOT a protein?

(1) Collagen

(2) Insulin

(3) Hemoglobin

(4) Inulin

33) Palmitic acid is an example of :–

(1) Essential amino acid.

(2) Non essential amino acid.

(3) Saturated fatty acid.

(4) Unsaturated fatty acid.

34) **Assertion :** Lipids are present in acid insoluble fraction during a chemical analysis of a tissue.
Reason : Lipids are macromolecules.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Assertion is True but the Reason is False.

(3) Both Assertion & Reason are False.

(4) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

35) Which of the following statements is **correct** ?

(1) Arachidonic acid has 20 carbon atoms excluding carboxyl carbon.

(2) In solutions of different pHs, the structure of amino acid changes.

(3) In a polysaccharide chain, left end is called reducing end and right end is called non-reducing end.

(4) Cellulose forms helical secondary structure.

36) Match the Column I & Column II.

Column - I		**Column - II**	
(A)	Cellulose	(i)	Exoskeleton of arthropods
(B)	Starch	(ii)	Does not contain complex helices & can not hold I_2

(C)	Chitin	(iii)	With I_2 gives blue colour
(D)	Glycogen	(iv)	Store house of energy in animal

(1) A – ii, B – iii, C – i, D – iv

(2) A – iii, B – ii, C – i, D – iv

(3) A – iii, B – ii, C – iv, D – i

(4) A – ii, B – iii, C – iv, D – i

37) **Statement-I :** In a polypeptide or a protein, amino acids are linked by a glycosidic bond.
Statement-II : A nucleotide has three chemically distinct components one is a heterocyclic compound, the second is a monosaccharide and the third a phosphoric acid or phosphate.

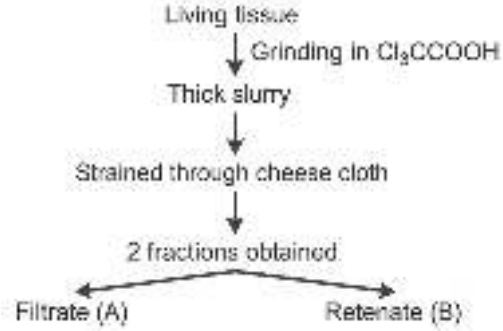
(1) Statement I and II both are correct.

(2) Statement I and II both are incorrect.

(3) Only Statement I is correct.

(4) Only Statement II is correct.

38) Consider this flow chart and a list of properties given after that :-



(i) Molecular weight ranging from 18-800 daltons approx
(ii) Proteins, nucleic acids, polysaccharides, and lipids
(iii) Chemicals more than 800 daltons molecular weight
(iv) Has monomers
(v) It generally has polymers.

(1) A : i, ii, iii and B : iv, v

(2) A : ii, iv and B : i, iii, v

(3) A : i, iv and B : ii, iii, v

(4) A : i, iii, v and B : ii, iv

39)

Name the class of enzyme that usually catalyze the following reaction :
$S - G + S^{\#} \rightarrow S + S^{\#} - G$
Where, G → Hydrogen
S → a substrate
$S^{\#}$ → another substrate

(1) Oxidoreductase

(2) Lyase

(3) Transferase

(4) Ligase

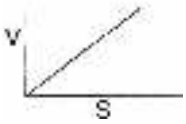
40) The complete catalytically active enzyme with its non-protein part is known as :

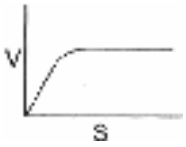
(1) Coenzyme
(2) Apoenzyme
(3) Holoenzyme
(4) Prosthetic group

41) Which one of the following graphs shows the relationship between the velocity of an enzymatic activity (V) and substrate concentration (S).

(1) 

(2) 

(3) 

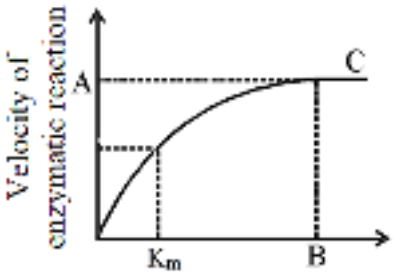
(4) 

42) **(A)** Metal ions are co-factors which form coordination bonds with side chains at the active site and at the same time form one or more coordination bonds with the substrate
(B) Co-enzymes are organic compounds and their association with the apoenzyme is permanent.

(1) Both (A) and (B) are correct
(2) Both (A) and (B) are incorrect
(3) Only (A) is correct
(4) Only (B) is correct

43) Which of the following statement is **incorrect** about enzymes ?

(1) Ribozyme is a non proteinaceous enzyme.
(2) Amount of activation energy decrease in the presence of enzymes.
(3) Enzyme is converted into product during reaction.
(4) NAD^+ and $NADP^+$ are examples of coenzyme.

44) 

Which one of the following is a **incorrect** conclusion with respect to above graph ?

(1) Beyond 'B' enzyme molecules are fewer than substrate molecules

(2) V_{max} is shown by 'A'

(3) Beyond 'B' there will be saturation of substrates

(4) K_m can be denoted with reference to 1/2 of V_{max}

45) Read the given passage and answer the following question :
'E' is a biological catalyst that increases the rate of biochemical reactions without being consumed. It has a specific three-dimensional structure with an active site where the substrate binds. The enzyme first forms a temporary complex with the substrate. In some cases, enzyme activity is regulated by inhibitors. Competitive inhibitors resemble the substrate and bind to the active site, whereas non-competitive inhibitors bind at a different site.
Select the correct sequence of steps involved in enzyme action:
W – Formation of enzyme-substrate complex
X – Binding of substrate to active site
Y – Release of product
Z – Formation of enzyme-product complex

(1) X → W → Z → Y

(2) W → X → Y → Z

(3) X → Z → W → Y

(4) Y → X → W → Z

46) Who showed that sunlight is essential for photosynthesis process ?

(1) Joseph priestley

(2) Jan Ingenhousz

(3) T.W. engelmann

(4) Van Niel

47) CO_2 fixation enzyme in bundle sheath cells in C_4 plants is.

(1) RuBP carboxylase

(2) PEP carboxylase

(3) Pyruvate carboxylate

(4) Carboxypeptidase

48) In light reaction, plastocyanin facilitates the transfer of electrons from :-

(1) PS-I to ATP synthase

(2) $Cytb_6f$ complex to PS-I

(3) PS-I to $NADP^+$

(4) PS-II to $Cytb_6f$ complex

49) Which of the following is major limiting factor in photosynthesis ?

(1) CO_2

(2) H_2O

(3) Light

(4) Temperature

50) Which of the following statement is **incorrect** ?

(1) Jan Ingenhousz showed that it is only the green part of the plants that could release oxygen.

(2) Most of the photosynthesis takes place in the blue and red regions of the spectrum.

(3) The light reaction of photosynthesis is directly light driven process.

(4) The products of dark reaction of photosynthesis are ATP and NADPH.

51) **Assertion (A) :** Cyclic flow of electrons results only in the synthesis of ATP, but not of $NADPH+H^+$.
Reason (R) : The stroma lamellae membranes lack PSII as well as NADP reductase enzyme and the excited electron does not pass on to NADP+ but is cycled back to the PSI.
Choose the correct answer from the options given below :

(1) (A) is correct but (R) is incorrect

(2) (A) is incorrect but (R) is correct

(3) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

52) **Statement-I :-** The relative rates of photosynthesis at different wavelengths of light is called action spectrum.
Statement-II :- The action spectrum of photosynthesis resembles roughly the absorption spectra of chlorophyll a and b.

(1) Statement I and statement II, both are correct

(2) Statement I and statement II, both are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

53) Match List I with List II

	List I		**List II**
(A)	Hatch and Slack pathway	(I)	Synthesis of ATP and $NADPH + H^+$

(B)	Photochemical phase of photosynthesis	(II)	3-Phosphoglyceric acid
(C)	Glycolate pathway	(III)	Oxaloacetic acid
(D)	C_3 pathway	(IV)	Phosphoglycolate

Choose the correct answer from the options given below :

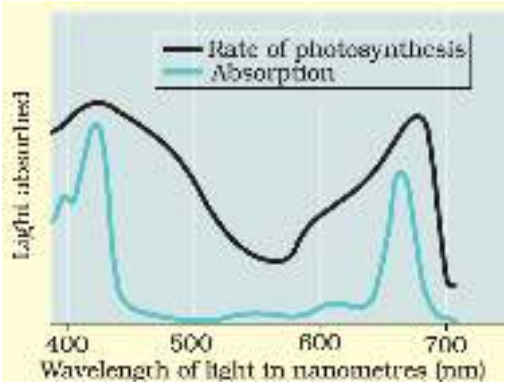
(1) A-III, B-I, C-IV, D-II

(2) A-I, B-II, C-III, D-IV

(3) A-III, B-II, C-I, D-IV

(4) A-IV, B-III, C-I, D-II

54) Which statement is **true** for given graph ?



(1) Minimum photosynthesis occur between region 500-600 nm

(2) Action spectrum of photosynthesis superimposed on absorption spectrum of all pigments

(3) Complete one to one overlap between the absorption spectrum of chl-a and the action spectrum of photosynthesis

(4) Rate of photosynthesis remains constant between region 400-700 nm

55) If 5 CO_2 enter into the Calvin cycle then what will be the difference in number of ATPs utilise in reduction and regeneration steps ?

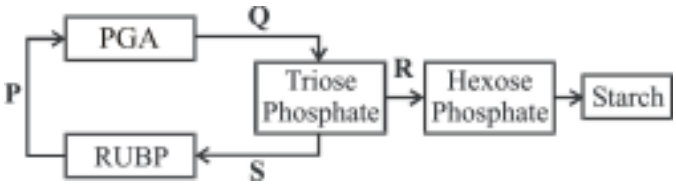
(1) Five

(2) Six

(3) Ten

(4) Eighteen

56) The diagram represents the Calvin cycle :-



At which stage CO_2 incorporated ?

(1) P

(2) Q

(3) R

(4) S

57) Primary carboxylation in C_3 plants and secondary carboxylation in C_4 plants occur with the help of _____&_____ respectively

(1) PEPcase & RuBisCO

(2) RuBisCO & RuBisCO

(3) RuBisCO & PEPcase

(4) PEPcase & ATPase

58) Fill in the blanks and choose the correct option

	Column-I		**Column-II**
(a)	Photosystem II	(I)	Reaction centre absorbs at .....(i)..... nm
(b)	Photooxidation	(II)	Prevented by .....(ii).....
(c)	H-carrier in Noncyclic photophosphorylation	(III)	Passes electron to .....(iii).....
(d)	PS I reaction centre	(IV)	Absorption peak at .....(iv).....

(1) i - 700 nm; ii - O_3 ; iii - Phytochrome ; iv - 680 nm

(2) i - 680 nm; ii - Accessory pigments ; iii - Cytochrome ; iv - 700 nm

(3) i - 680 nm; ii - Chl 'a' ; iii - Phytochrome ; iv - 700 nm

(4) i - 700 nm; ii - Main pigment ; iii - Chl 'b' ; iv - less then 700

59) While studying the vertical section of a leaf anatomy under a high power microscope a researcher observed a specific group of large cells surrounding the vascular bundles. These cells were characterised by having thick walls impervious to gaseous exchange and totally lacked intercellular spaces. Upon staining they showed a very high number of chloroplast.
Which of the following event is strictly restricted to the specific cells observed by the researcher ?

(1) The initial fixation of atmospheric CO_2 by PEP carboxylase.

(2) The synthesis of ATP and NADPH through non cyclic photophosphorylation

(3) The decarboxylation of C_4 acids to release CO_2.

(4) The transport of C_4 acids like malic acid or aspartic acid out of the cell.

60) In the following flow diagram, identify the correct events of C_4 plant during dark reaction of photosynthesis.
[CO_2 + PEP] → [1] → [2] → [3] → [4] → [5] → [Sugar]
I. C_4 acid belongs to category 1
II. Malic acid or Aspartic acid belong to category 2
III. Transportation of Malic acid from mesophyll cell to Bundle sheath cells belongs to category 3
IV. Oxidative decarboxylation of malic acid in bundle sheath cell belong to category 4
V. CO_2 + RuBP forms Sugar via C_3 Cycle belongs to category 5

(1) I, II, III and IV only

(2) I, II, IV only

(3) I, II, V only
(4) I, II, III, IV and V all

61) Product of Kreb's cycle is :

(1) $NADPH_2$
(2) $NADH_2$
(3) O_2
(4) Pyruvic acid

62) Which of the following step of glycolysis shows the reduction of NAD^+ ?

(1) Glucose-6 PO_4 → Fructose-6-PO_4
(2) 3-PGAL ⇌ DHAP
(3) 1,3-bis PGA → 3 PGA
(4) 3PGAL → 1,3-bis PGA

63) Cytochrome C oxidase complex of mitochondrial ETS contains :

(1) Cytochrome a, a_3 and one copper centre
(2) Cytochrome a and two copper centres
(3) Cytochrome a_3 and one copper centre
(4) Cytochrome a, a_3 and two copper centre

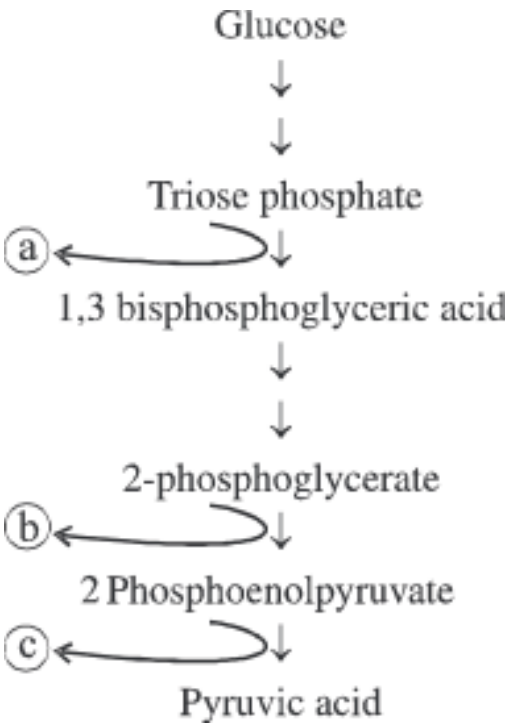
64) All of the following processes yield NAD^+, except :-

(1) Lactic acid fermentation
(2) Alcohol fermentation
(3) Glycolysis
(4) ETS and oxidative phosphorylation

65) At how many places in the kreb's cycle NAD^+ reduced to $NADH+H^+$ and FAD^+ reduced to $FADH_2$, respectively:

(1) One and three
(2) Three and one
(3) Four and one
(4) Three and two

66) Find out the correct matching :-



(1) **a**-ATP, **b**-NADH, **c**-H_2O

(2) **a**-H_2O, **b**-NADH, **c**-ATP

(3) **a**-NADH, **b**-H_2O, **c**-ATP

(4) **a**-H_2O, **b**-ATP, **c**-NADH

67) **Assertion :** Cellular respiration is an amphibolic pathway
Reason : The carbon skeleton produced during breakdown of respiratory substrates is used as precursor for biosynthesis of other molecules in cell.
In the light of the above statements, choose correct answer from options given below :

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

68) Which of the following is **not** correct regarding cellular respiration ?

(1) In both lactic acid and alcohol fermentation not much energy is released

(2) The respiratory quotient depends upon the type of respiratory substrate used during respiration.

(3) All the pathways of the process work simultaneously

(4) Only glucose is being respired, no other alternative substrates are entering in the pathway at any of the intermediary stages.

69) Which among the given enzymes are responsible for conversion of glucose to ethanol and CO_2 ?
Hexokinase, Pyruvic acid decarboxylase, Citrate synthase, Alcohol dehydrogenase, Lactate dehydrogenase.

(1) 2

(2) 3

(3) 4

(4) 5

70) Match List I with List II :

List I		List II	
A	TCA cycle	I	Cytoplasm
B	Oxidative phosphorylation	II	Inner membrane of mitochondria
C	Glycolysis	III	Mitochondrial matrix
D	Oxidative decarboxylation	IV	Citrate synthase

(1) A-IV,B-II,C-I,D-III

(2) A-IV,B-I,C-III,D-II

(3) A-I,B-III,C-II,D-IV

(4) A-II,B-IV,C-III,D-I

71) **Statement-I :-** In TCA cycle, two successive steps of decarboxylation lead to the formation of α-keto glutaric acid and then succinyl-CoA.
Statement-II :- During the conversion of succinyl-CoA to succinic acid a molecule of GTP is synthesised.

(1) Statement I is correct but statement II is incorrect

(2) Both statements are incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statements are correct

72) What is the **correct sequence** of e^- transfer when $FADH_2$ is oxidised in ETS ?

(1) FMN → UQ → Cyto-c → Cyto-b

(2) FMN → Cyto-b → Cyto-c → Cyto-c_1

(3) Cyto-b → Cyto-a_3 →Cyto-c_1 → Cyto-c

(4) Cyto-b → Cyto-c_1 → Cyto-c → Cyto-a

73) Consider the reactions and arrange the correct sequence with respect to glycolysis :
i) Formation of fructose - 1,6-bisphosphate
ii) Dehydrogenation
iii) Cleavage to form triose-3-phosphate
iv) Second substrate-level phosphorylation
v) Dehydration

(1) i → ii → iii → iv → v

(2) i → iii → ii → v → iv

(3) i → ii → iii → v → iv

(4) i → iii → ii → iv → v

74) **Statement-I :** The initial process of cellular respiration takes place in the cytoplasm in which glucose molecule is broken down into pyruvic acid and CO_2. The fate of the pyruvate depends on the availability of oxygen, the organism and cellular need.
Statement-II : The energy, in the NADH + H^+ and $FADH_2$ generated in the matrix of the mitochondria is used to synthesised ATP. When electron form these molecules move through ETS, they release enough energy to synthesis ATP. This is called oxidative phosphorylation.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

75) A biotechnology company is optimizing its industrial ethanol production process using yeast fermentation. During fermentation under anaerobic conditions, yeast converts glucose into ethanol, carbon dioxide, and other minor acidic byproducts. The process is monitored closely for pH changes, gas release, and substrate consumption. Despite controlled temperature and nutrient supply, occasional decreases in ethanol yield were observed ?
Which of the following explanations best aligns with the observations in this fermentation system?

(1) Contamination by aerobic bacteria caused a shift to lactic acid fermentation, increasing alkalinity and reducing ethanol yield.

(2) Yeast switched to aerobic respiration due to oxygen contamination, producing water and carbon dioxide instead of ethanol.

(3) The glucose substrate underwent incomplete oxidation producing fatty acids and lowering carbon dioxide output.

(4) Yeast increased ethanol production under stress, causing heightened carbon dioxide and less acidic medium.

76) Which of the following hormone causes rapid elongation of internode in deep water rice plant so that leaves and upper part of shoot to remain above water ?

(1) Cytokinin

(2) Ethylene

(3) ABA

(4) NAA

77) Which hormone is used to induce rooting from cut end of the stem-

(1) GA_1

(2) Auxin

(3) ABA

(4) Kinetin

78) Which of the following induces femaleness in cucumber?

(1) Ethylene

(2) Ethanol

(3) ABA

(4) Gibberellin

79) Under certain conditions regaining the lost capacity of division by living cells is called:

(1) Dedifferentiation

(2) Redifferentiation

(3) Differentiation

(4) Efficiency index

80) Coconut milk stimulate cell division as it is rich source of :-

(1) Auxin

(2) Cytokinin

(3) GA_3

(4) ABA

81) Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Plants retain the capacity for unlimited growth throughout their life.
Reason (R) : Generally growth is accompanied by metabolic processes that occur at the expense of energy.
In the light of the above statements, choose the **correct** answer from the options given below :

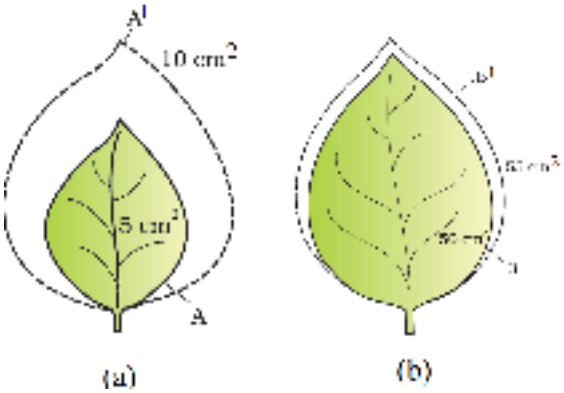
(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

82) What would be the **correct** option regarding given figure (a) and (b).



(1) Relative growth rate is equal in both figure.

(2) Figure (a) show more absolute growth rate as compared to figure (b).

(3) Absolute growth rate is equal in both figure (a) and (b).

(4) Relative growth rate is more in figure (b) as compared figure (a).

83) Which of the following statement is **incorrect** ?

(1) The first step in the process of plant growth is dedifferentiation

(2) Development of a mature plant from a fertilized egg follow highly ordered succession of events

(3) Plant growth and development is related to water status of plant

(4) Vegetative phase precede flowering in plants

84) Which one of the following options gives the **correct** categorisation of plant growth regulators (**A, B, C**) according to their physiological effect ?

	A **Auxin**	**B** **Abscisic Acid**	**C** **Ethylene**
(1)	Induce parthenocarpy	Seed maturation	Thinning in cotton
(2)	Malting process	Parthenocarpy	Weed free lawns
(3)	Bolting process	Stress hormone	Xylem differentiation
(4)	Respiratory Climactic	Apical dominance	Brewing industry

(1) 1

(2) 2

(3) 3

(4) 4

85) **Statement-I :** Development includes all changes that an organism goes through during it life cycle from germination of the seed to senescence.
Statement-II : Plasticity is the ability of plants to form different kinds of structures in response to environment or phase of life.

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

86)

How many plant events in the list given below are affected by the interaction of more than one PGR?

Dormancy in seeds, Senescence, Apical dominance, Abscission, Dormancy in buds

(1) Five

(2) Four

(3) Two

(4) Three

87)

Which of the following is **not** true regarding plasticity in plants ?

(1) In cotton, leaves of juvenile plant is similar to that of mature plant

(2) Heterophylly in butter cup is an example of plasticity

(3) Plants can show plasticity in response to environment

(4) Cotton, coriander, larkspur and butter cup show plasticity

88) Identify **a, b, c** and **d** in the given sentences.
I. Spraying sugarcane crop with ....**a**.... increases in the length of the stem, thus increasing the yield by as much as ....**b**.... tonnes per acre.
II. .... **c**.... does not occur naturally in plants.
III. Search for natural substances with cytokinins like activities led to the isolation of ...**d**... from corn kernels and coconut milk.

(1) **a**-auxins, **b**-10, **c**-NAA, **d**-zeatin

(2) **a**-gibberellins, **b**-20, **c**-zeatin, **d**-kinetin

(3) **a**-gibberellins, **b**-10, **c**-zeatin, **d**-kinetin

(4) **a**-gibberellins, **b**-20, **c**-kinetin, **d**-zeatin

89) The cells derived from root apical and shoot apical meristems and intrafasicicular cambium show a phenomenon, leading to maturation, called differentiation. Differentiation results in development of structures that is commensurate with the function the cells finally has to perform. Which of the following changes in plants represents this dependency of cellular function on the structure of cell, i.e. differentiation most appropriately ?

(1) Formation of secondary meristem cork cambium from fully differentiated parenchyma cells.

(2) Formation of secondary meristem interfascicular cambium from fully differentiated parenchyma cells.

(3) To form tracheary elements of primary xylem, cells lose protoplasm and develop a strong, elastic, lignocellulosic secondary cell walls.

(4) Cells produced from cork cambium lose the capacity to divide.

90) Match the column I, II and III :

Column-I		**Column-II**		**Column-III**	
(a)	Auxin	(i)	Carotenoids	(A)	Apical dominance
(b)	Cytokinins	(ii)	Indole compounds	(B)	Antagonistic to GA_3
(c)	Abscisic acid	(iii)	C_2H_4	(C)	Fruit ripening
(d)	Ethylene	(iv)	Adenine	(D)	Rapid cell division

Choose the correct answer from the options given below :-

(1) a-ii-A, b-i-B, c-iv-D, d-iii-C

(2) a-ii-A, b-iv-D, c-i-B, d-iii-C

(3) a-ii-A, b-iv-D, c-i-C, d-iii-B

(4) a-ii-D, b-iv-A, c-iii-C, d-i-B

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	4	2	3	4	2	3	2	3	4	1	2	3	2	3	3	4	3	3	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	2	4	1	3	3	2	2	3	3	3	2	1	3	2	1	1	4	3	3
Q.	41	42	43	44	45															
A.	1	4	2	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	1	1	2	3	2	3	1	3	1	3	2	3	2	2	3	4	3	1	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	3	2	3	1	1	3	1	2	2	4	1	4	2	2	3	3	2	1	4
Q.	86	87	88	89	90															
A.	1	1	1	1	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	2	3	4	2	4	2	4	1	4	3	3	4	2	3	1	3	3	4	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	1	1	1	4	1	3	3	1	2	3	4	3	2	2	1	4	3	1	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	3	3	1	2	1	2	1	4	3	1	1	1	1	1	2	2	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	4	4	3	2	3	1	4	2	1	4	4	2	3	2	2	2	1	1	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	3	1	1	3	1	1	4	3	2										

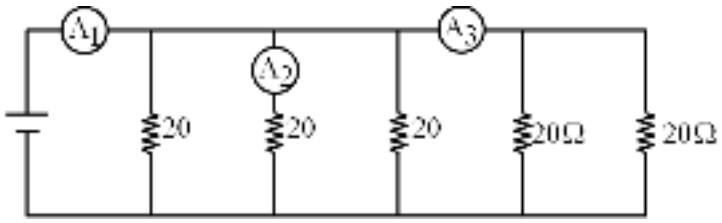
SOLUTIONS

PHYSICS

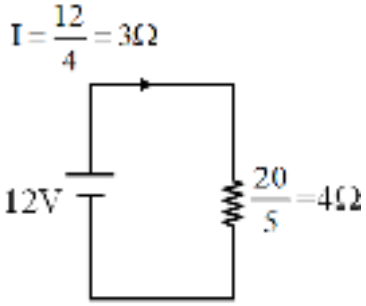
1) **Explain the question :** To find reading A_1, A_2 and A_3 as given figure.

Concept : This question is based on basic circuit analysis.

Solution :



↓ All resistance in parallel



$A_1 = \frac{V}{R_{eq}} = \frac{12}{4}$ = 3 amp

Current in each branch = $\frac{V}{R_{branch}} = \frac{12}{20} = \frac{3}{5}$ amp

So, $A_2 = \frac{3}{5}$ amp = 0.6 amp

and $A_3 = \frac{3}{5} + \frac{3}{5} = \frac{6}{5}$ = 1.2 amp

Final answer : Option (3)

2) For balanced WSB, no current flows in BD branch. So R_2 can be any finite value.

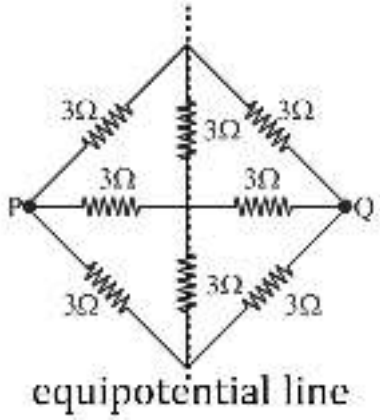
Also $\frac{10}{30} = \frac{R_1}{9} \Rightarrow R_1 = 3\Omega$

3)

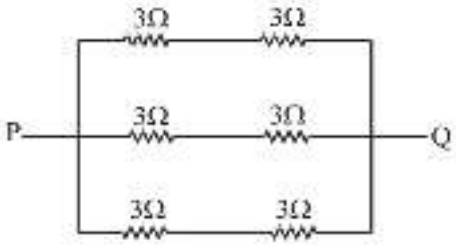
Explain Question : Find net resistance between P and Q.

Concept : This question is based on combination of resistance.

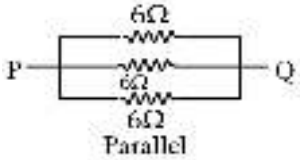
Solution :



All the resistors lies on the equipotential line will be useless they can be removed then circuit becomes



⇒ series



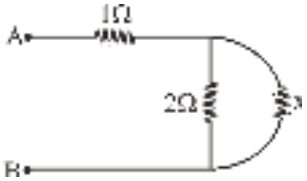
⇒

$$\frac{1}{R_{eq}} = \frac{1}{6} + \frac{1}{6} + \frac{1}{6}$$

$$R_{eq} = \frac{6}{3} = 2\Omega$$

Final answer : Option (2)

4) $x = 1 + \frac{2x}{2+x}$



$x^2 - x - 2 = 0$

$x^2 - 2x + x - 2 = 0$

$x(x-2) + 1(x-2) = 0$

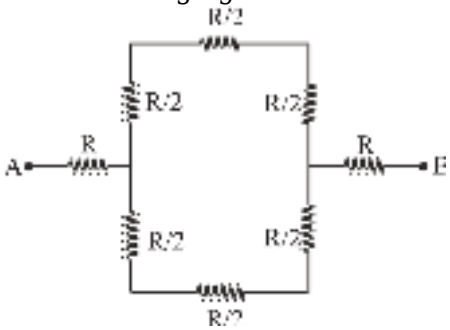
$x = -1 \;\; ; \;\; x = 2\Omega$

5) **Explain the Question :** Asking about the equivalent resistance between A & B.

Concept : This question is based on series & parallel only.

Solution :

Now rearranging the circuit.



$$R_{AB} = R + R + \frac{1}{2}\left(\frac{3R}{2}\right)$$

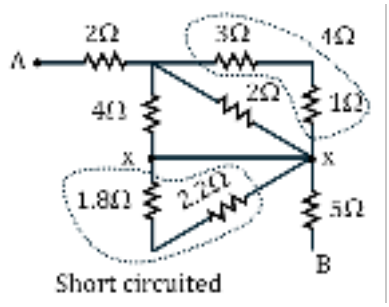
$$R_{AB} = 2R + \frac{3R}{4}$$

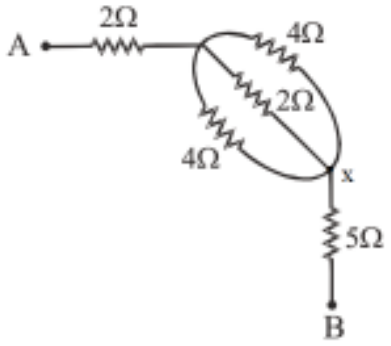
$$R_{AB} = \frac{11R}{4}$$

Final Answer : Option (4)

6)

Question Explanation : A complex resistor network is given between points A and B with resistors of values: 2Ω, 3Ω, 4Ω, 1Ω, 1.8Ω, 2.2Ω, and 5Ω. Use series-parallel reduction to find the equivalent resistance across A and B.
Concept : This question is based on Series-parallel resistor simplification
Formula : $R_{eq} = R_1 + (R_2 \,||\, R_3) + \ldots$
Solution :





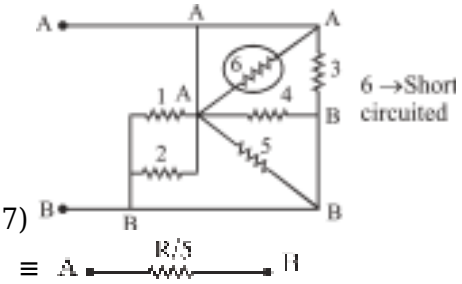
$$R_{AB} = 2 + \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{4}\right)^{-1} + 5$$

$$R_{AB} = (2) + (1) + (5)$$

$R_{eq} = 8\Omega$

Final answer : Option (2)

7)

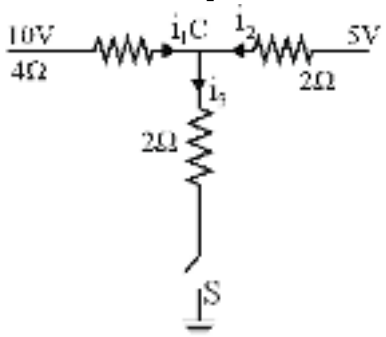


8) $I_1 = I_2 = 2$ Amp.

$V_A - 2(3) - 5 - 0(2) - 2(10) = V_B$

$V_A - V_B = 6 + 5 + 20 = 31$ volt

9) Let V be the potential at C.



Using Kirchhoff's first law, $i_1 + i_2 = i_3$

$$\frac{10-V}{4} + \frac{5-V}{2} = \frac{V-0}{2}$$

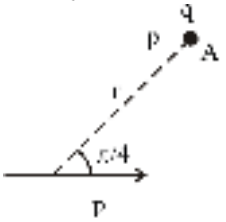
On solving, V = 4V, $i_3 = V/2 = 4/2 = 2A$

10)

$V_A - 6 - 3 - 4 + 2 - 2 = V_B$

$V_A - V_B = +13$ volt

11) **Question Explanation:**



We have to calculate force on charge 'q' placed at point 'A' due to dipole

Concept: (i) electric field at arbitrary point due to dipole

(ii) F = qE

Solution:

$$E_P = \frac{KP}{r^3}\sqrt{1 + 3\cos^2 45^\circ}$$

$$E_P = \frac{KP}{r^3}\sqrt{1 + 3 \times \frac{1}{2}}$$

$$E_P = \frac{KP}{r^3}\sqrt{\frac{5}{2}}$$

$$|F| = q\,|E_P| = q\frac{KP}{r^3}\sqrt{\frac{5}{2}}$$

$$|F| = \sqrt{\frac{5}{2}}\frac{KPq}{r^3}$$

Final Answer: Option (1)

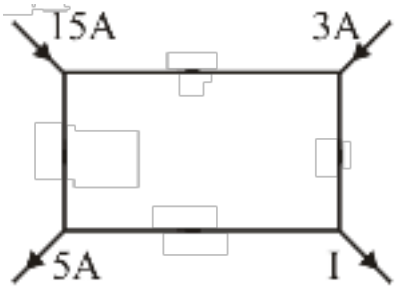
12)

Explain Question : We have to find I as shown in figure.

Concept : This question is based on Kirchhoff's law.

Solution : By using KCC

$I_{in} = I_{out}$

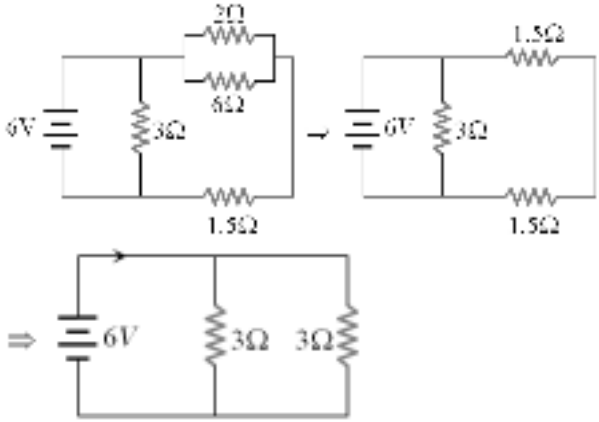


15 + 3 – 5 = I

I = 13A

Final answer : Option (2)

13) Given circuit can be redrawn as follows



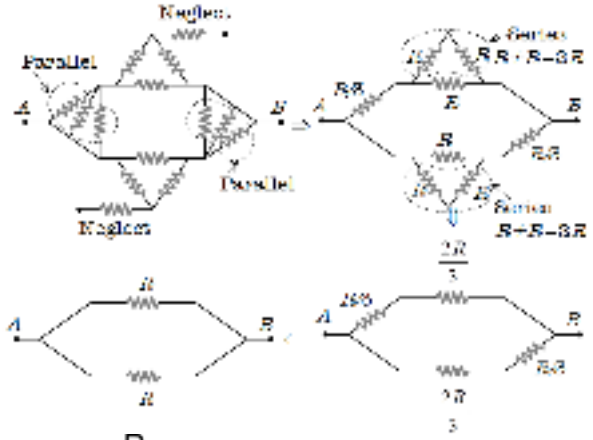
$$\Rightarrow i = \frac{6}{1.5} = 4\,A$$

14) $\text{Volume} = A\ell = 3 \Rightarrow A = \frac{3}{\ell}$

Now $R = \rho\frac{\ell}{A} \Rightarrow 3 = \frac{\rho \times \ell}{3/\ell} = \frac{\rho\ell^2}{3}$

$$\Rightarrow \ell^2 = \frac{9}{\rho} = \frac{3}{\sqrt{\rho}}$$

15) Given circuit can be redrawn as follows



$$\Rightarrow R_{eq} = \frac{R}{2}$$

16) When charge q is released in uniform electric field E then its acceleration $a = \frac{qE}{m}$ (is constant)

So its motion will be uniformly accelerated motion and its velocity after time t is given by

$v = at = \frac{qE}{m}t$

$\Rightarrow KE = \frac{1}{2}mv^2 = \frac{1}{2}\left(\frac{qE}{m}t\right)^2 = \frac{q^2E^2t^2}{2m}$

17) Dipole moment $p = 4 \times 10^{-8} \times 2 \times 10^{-4} = 8 \times 10^{-12}$ m

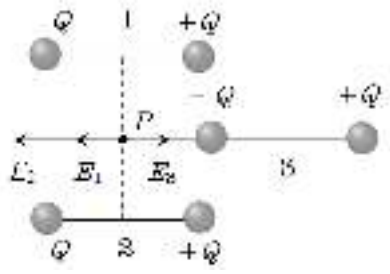
Maximum torque = $pE = 8 \times 10^{-12} \times 4 \times 10^{8}$

$= 32 \times 10^{-4}$ Nm

Work done in rotating through $180^{o} = 2pE$

$= 2 \times 32 \times 10^{-4} = 64 \times 10^{-4}$ J

18)



Point P lies at equatorial positions of dipole 1 and 2 and axial position of dipole 3.

Hence field at P

due to dipole 1

$E_1 = \frac{k.p}{x^3}$ (towards left)

due to dipole 2

$E_2 = \frac{k.p}{x^2}$ (towards left)

due to dipole 3 $E_3 = \frac{k.(2p)}{x^3}$ (towards right)

So net field at P will be zero.

19)

(A) 5 (2 – 0) = 10

(B) $\left[\frac{x^2}{2}\right]_1^2 = \frac{4-1}{2} = 1.5$

(C) $[\ln x]_1^2 = \ln\left(\frac{2}{1}\right) = \ln 2$

(D) $\frac{[e^{-x}]_0^{\infty}}{-1} = e^0 - e^{-\infty} = 1 - 0$

$= 1$

20)

$y = kx^2$, parabola opening upwards

21) $\frac{dy}{dx} = \frac{e^x(1) - (1+x)e^x}{(e^x)^2}$

$= \frac{e^x[1-1-x]}{e^x.e^x} = \frac{-x}{e^x}$

22) $\int_3^5 \frac{1}{2x+3}dx = \left[\frac{\ell n(2x+3)}{2}\right]_3^5$

$= \frac{1}{2}[\ln(13) - \ln(9)]$

$= \frac{1}{2}\ell n\left(\frac{13}{9}\right)$

23)

3x + 4y + 1 = 0

x = 0; y = –1/4

y = 0; x = –1/3

both intercept -ve

& slope -ve

so option (4)

24) $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

$P^2(10) = 4P^2 + 2P^2 + 4\sqrt{2}\,P^2 \cos\theta$

$\cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45°$

25) $\left|\vec{A}+\vec{B}\right| = \left|\vec{A}-\vec{B}\right| \Rightarrow \sqrt{A^2+B^2+2AB\cos\theta} = \sqrt{A^2+B^2-2AB\cos\theta}$

$\Rightarrow 4AB\cos\theta = 0$

$\cos\theta = 0 \Rightarrow \theta = 90°$

26)

Change in velocity = $2v\sin(\theta/2) = 2v\sin 20°$

27) Note that $\vec{B} \times \vec{A} \perp \vec{A}$. Hence their dot product is zero.

28) $\cos\theta = \frac{\vec{A}\cdot\vec{B}}{AB}$

Here $\vec{A}\cdot\vec{B} = -2+6-4 = 0$

cos $\cos\theta = 0 \Rightarrow \theta = 90°$

29) **Question Explanation:**
Analyzing the reflection of a pressure pulse (longitudinal wave) at the open and closed boundaries of a pipe.
Concept:
Reflection of longitudinal waves at boundaries.
Closed End (Rigid Boundary): A pressure pulse reflects without a phase change in pressure. A high-pressure pulse reflects as a high-pressure pulse (compression reflects as compression).
Open End (Free Boundary): The pressure at the open end must remain atmospheric. To maintain this, a pressure pulse reflects with a 180° phase change. A high-pressure pulse reflects as a low-pressure pulse (compression reflects as rarefaction).
Solution:
At an open end: A high-pressure pulse is reflected as a low-pressure pulse. (Statement B is correct).
At a closed end: A high-pressure pulse is reflected as a high-pressure pulse. (Statement D is correct).
Final Answer:
Option (3)

30) **Question Explanation:** Find a fork's frequency using a sonometer.

Concept: Sonometer & Beat Frequency

Solution: Let the frequency of the tuning fork be **'f'**. The frequency of a sonometer wire (f_{wire}) is inversely proportional to its length (L), so f_{wire}·L is constant. The shorter wire (20 cm) has a higher frequency, f1, and the longer wire (21 cm) has a lower frequency, f_2. Since the beat frequency (5 Hz) is the same in both cases, the fork's frequency **'f'** must lie between the two

wire frequencies.

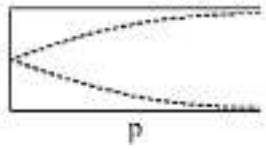
Therefore, $f_1=f+5$ and $f_2=f-5$.
Using the relation $f_1L_1=f_2L_2$:
$(f+5)\times20=(f-5)\times21$
$20f+100=21f-105$
$205=f$
$\Rightarrow f = 205$ Hz

Final Answer:(3)

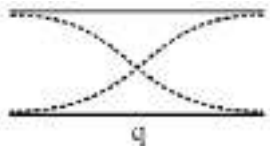
31) $\frac{V}{4L_1} = 3\frac{V}{2L_2}$
$\frac{L_1}{L_2} = \frac{1}{6}$

32)

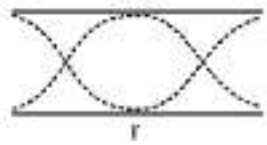
Explain the Question : Four air columns of equal length show different standing-wave patterns p, q, r, s. From the number of nodes/antinodes in each pattern, we must compare their frequencies.
Concept : This question is based on standing waves in air columns (open and closed pipes).
Formula : Frequency ∝ number of segments (harmonic order) for the same length.
Solution :
From the figures:



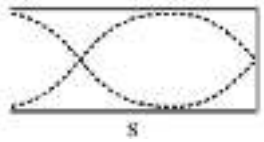
p shows the fundamental mode → $n_p = \frac{V}{4L}$,
lowest frequency (1 unit).



q shows the next higher mode with twice the segments →
$n_q = \frac{V}{2L}$
frequency = 2 units.



r shows four segments →
$n_r = \frac{2V}{2L}$
frequency = 4 units.



s shows three segments →

$n_s = \frac{3V}{4L}$

frequency = 3 units.

Thus,

$n_p : n_q : n_r : n_s = 1 : 2 : 4 : 3$

Final answer : 2

33) $\frac{\lambda}{4} = \ell_1 + e \& \frac{3\lambda}{4} = \ell_2 + e \Rightarrow \lambda = 2(\ell_2 - \ell_1)$

Speed of sound

$= n\lambda = 512 \times 2 \times \left(34 \times 10^{-2}\right) = 348$ m/s

Given $y = a + bt + ct^2 - dt^4$

$\therefore \upsilon = \frac{dy}{dt} = 0 + b + 2ct - 4dt^3$

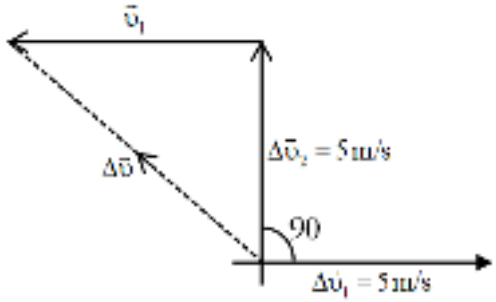
Putting t = 0, $\upsilon_{initial}$ = b so initial velocity = b

Now, acceleration

(a) $= \frac{d\upsilon}{dt} = 0 + 2c - 12dt^2$

Putting t = 0, $a_{initial}$ = 2c

34)



35)

$\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$

$\Delta v = \sqrt{v_1^2 + v_2^2 - 2v_1 v_2 \cos 90^\circ} = \sqrt{5^2 + 5^2} = 5\sqrt{2}$

$\Delta v = 5\sqrt{2}$

Average acceleration

$= \frac{\Delta v}{\Delta t} = \frac{5\sqrt{2}}{10} = \frac{1}{\sqrt{2}} m/s^2$

towards north-west (As clear from the figure).

36) $s = t^3 - 3t^2 + 2$

$\frac{ds}{dt} = 3t^2 - 6t$

and acceleration $\frac{d^2s}{dt^2} - 6t - 6$

when acceleration is zero, t = 1 sec.

Hence, displacement of the particle at t = 1 sec (i.e., when acceleration becomes zero) is,

$s = t^3 - 3t^2 + 2$

$=1 - 3 + 2 = 0m.$

37) by using $S_n = u + \frac{a}{2}(2n-1).$ Distance travelled by body A is 5th second

$= 0 + \frac{a_1}{2}(2 \times 5 - 1)$

Distance travelled by body B is 3rd second is

$= 0 + \frac{a_2}{2}(2 \times 3 - 1)$

According to problem : $0 + \frac{a_1}{2}(2 \times 5 - 1)$

$= 0 + \frac{a_2}{2}(2 \times 3 - 1)$

$\Rightarrow 9a_1 = 5a_2 \Rightarrow \frac{a_1}{a_2} = \frac{5}{9}$

38) $s_n = \frac{1}{2}a\ n^2$

$s_{nth} = \frac{1}{2}a(2n-1)$

$\frac{s_n}{s_{nth}} = \frac{n^2}{(2n-1)}$

39)

A body thrown vertically upwards under gravity then at heighest point its velocity is zero but acceleration is non zero.

40) $S = \frac{V^2}{2a}$

$\frac{S_1}{S_2} = \left(\frac{V_1}{V_2}\right)^2$

$\frac{4}{S_2} = \left(\frac{40}{100}\right)^2 = \frac{0.16}{1}$

$S_2 = \frac{4}{0.16} = \frac{400}{16} = 25m$

41) Ans. (1)

$S = 2r \sin\frac{\theta}{2}$

$= 2(2) \sin\left(\frac{\pi}{3}\right)$

$= 2\sqrt{3}\text{m}$

42) **Explain the Question :** The user wants to identify the correct relationship between the velocity and acceleration of a body in one-dimensional motion and how it affect its speed.
Concept : This question is based on speed changes based on the relative directions of velocity and acceleration.
Solution :
Speed increases when velocity and acceleration have the same sign. Speed decreases when they have opposite signs.

When velocity is negative and acceleration is positive, they are in opposite directions, causing the body to slow down.
Given $v<0$ and $a>0$, the vectors for velocity and acceleration point in opposite directions. This opposition causes the magnitude of the velocity, which is speed, to decrease.
Final answer : 4

43) **Question Explanation:**
Calculate the ratio of the distance covered by a vertically projected body during its first second to that during its seventh second, given it takes 6 seconds to reach maximum height.

Concept:

Distance in n^{th} second: $S_n = u + \frac{a}{2}(2n-1)$. For downward motion from rest, distance is $S_n = \frac{1}{2}g(2n-1)$.

Solution:
Given time to reach maximum height is t = 6 s, then $v = u - gt \Rightarrow 0 = u - 10(6) \Rightarrow u = 60\ \text{m/s}$.
1. Distance in 1st second (upward):
$S_1 = u - \frac{g}{2}(2(1)-1) = 60 - 5(1) = 55\ \text{m}$
2. Distance in 7th second (downward):
The body reaches the top at t = 6 s and falls. The 7th second of total motion corresponds to the 1st second of its downward journey:
$S_7 = 0 + \frac{g}{2}(2(1)-1) = 5\ \text{m}$
3. Ratio:
$\text{Ratio} = \frac{S_1}{S_7} = \frac{55}{5} = \frac{11}{1}$

Final Answer: Option(2)

44)

$S = S_1 - S_2 = \frac{1}{2}g(3)^2 - \frac{1}{2}g(2)^2$
$S = \frac{1}{2} \times 10[9-4] = 25\text{m}$

45) **Question Explanation :** A body is projected vertically upward from the ground. We need to select the correct velocity-time (V−T) graph for its entire flight until it returns to the ground.

Concept: Velocity-time graph for an object under free fall.

Formula:

$v = u + at$

$y = mx + c$

Solution:

Given:

A body is projected vertically upward, under freefall

Let the initial velocity = u

final velocity = v

Acceleration due to gravity = –g (– sign is showing the direction)

time = t

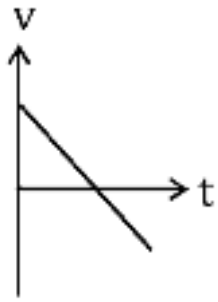
 $v = u – gt$ or $v = –gt + u$

comparing with $y = mx + c$ (straight line equation)

We can see that lope m = –g (negative)

Intercept, C = u

So the correct graph for this condition is-



Hence, option (2) is correct.

Final Answer : Option (2)

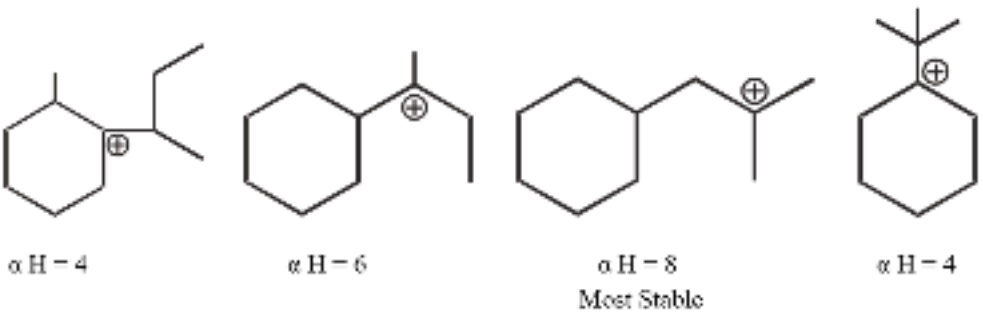
CHEMISTRY

46)

Question Explanation: We need to determine which carbocation among the given cyclohexane derivatives is most stable.

Concept: Carbocation stability, hyperconjugation, inductive effect, alkyl substitution.

Solution:



Final Answer: Option (3)

47)

Stability of carbanion ∝ EWG ∝ $\frac{1}{ERG}$

48)

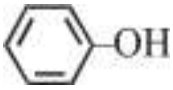
NCERT XI Hydrocarbon



49)
• cyclic
• planar
• cyclic resonance
• follow Huckel's Rule
[(4n + p) e^- system; n = 0, 1, 2, ---]
4n + 2 = 6
4n = 4 : $\boxed{n = 1}$
• Aromatic

50)

NCERT XII /Chapter 7 / Table 7.3

51) **Question Explanation:** The question asks which has the least bond length in the C-O bond.
Concept: This question is based on bond length of C – O is influnced by resonance.
Solution :-


Here –OH is in conjugation with ring. Therefore partial double bond character comes i.e. bond length becomes least.
Final Answer: 2.

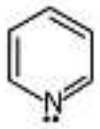
52)

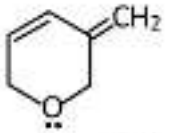
NCERT / XII / Amine

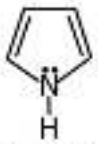
53) **Question Explanation**: Ordering the stability of resonating structures.
Concept: Resonance stability
Solution: The stability of a resonance structure is determined by the octet rule, charge separation, and electronegativity of charged atoms. The structure with no charge separation and all atoms having a complete octet is the most stable. The correct order is I > II > III.

54) **Question Explanation :** Identify compound in which lone pair is involved in Resonance.

Concept : Resonance Effect.
Solution :

(I)  (localised ℓ.P)

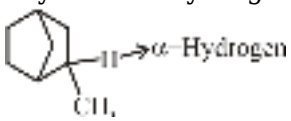
(II)  (localised ℓ.P)

(III)  (Delocalised lone pair)

(IV) $CH_2=CH-\ddot{O}-CH_3$ (Delocalised lone pair)

Final Answer: Option (3)

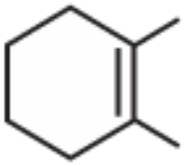
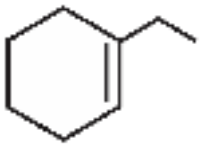
55) Only C → α - Hydrogen



A & B α Hydrogen absent

56)

Explaining the Question :- The question asks to arrange in order of stability

(I)  (II)  (III) 

Concept:- This question is based on Hyperconjugation
Solution :- More hyperconjugated an alkene is more is its stability
More the count of α-Hydrogen more is stability of an alkene, thus in structures given (I) has 4 α-Hydrogen (II) has 10 α-
Hydrogen and (III) has 6 α-Hydrogen atoms thus order of stability is (II) > (III) > (I).
Increasing order is I < III < II
Final Answer :- (3)

57)

A.S. $\propto \frac{-m}{+m} \propto \frac{-H}{+H} \propto \frac{-I}{+I}$

58)

NCERT / XI / 8.7.6

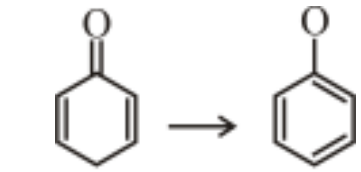
59)

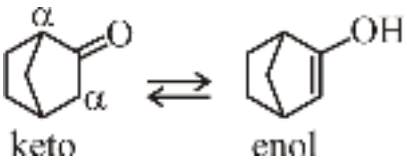
Basic strength $\propto \frac{+M}{-M} \propto \frac{+H}{-H} \propto \frac{+I}{-I} \propto \frac{1}{\text{EN of nitrogen}}$

60)

NCERT / XI / 8.7.7

61)





62)

Question Explanation: Determine whether the **Assertion** and **Reason** are correct and whether the Reason explains the Assertion.

Concept: Tautomerism, Aromaticity, Resonance, Enol Stability

Solution: Resorcinol **enol form is more stable** because it retains **aromaticity** of the benzene ring. The keto form loses **aromatic character**, making it less stable. Hence, **aromaticity directly explains the greater stability of the enol form**.

Assertion and Reason are true and Reason is the correct explanation.

Final Answer: Option (4)

63)

$P_S = P_A^o X_A + P_B^o X_B$

$= 45 \times \frac{1}{3} + 36 \times \frac{2}{3}$

$= \frac{45+72}{3} = \frac{117}{3} = 39$

$P_{S_{obs}} > P_{S_{Cal}}$ +ve deviation

64) $m = \frac{c \times 1000}{M_B \times (a-c)}$

m = b

$$M_B = \frac{m \times (a-c)}{c \times 1000} = \frac{b(a-c)}{c \times 1000}$$
$$M_B = \frac{c \times 1000}{b(a-c)}$$

65)

X is latent heat of vaporisation
ΔH_{vap} = enthalpy of vap (cal/mol)

66) $$Y_P = \frac{P_P^o X_P}{P_P^o X_P + P_h^o X_h} \quad \text{... (i)}$$
$$Y_h = \frac{P_h^o X_h}{P_P^o X_P + P_h^o X_h} \quad \text{... (i)}$$
$$\frac{\text{(i)}}{\text{(ii)}}$$
$$\frac{Y_P}{1-Y_P} = \frac{440}{120} \cdot \frac{1}{4} = \frac{11}{12}$$
$11 - 11Y_P = 12Y_P$
$$Y_P = \frac{11}{23} = 0.478$$

67) $\Delta T_f = i\, K_f \times m$
$$2.8 = 1 \times 1.86 \times \frac{w}{62 \times 1}$$
w = 93.3 gm

68) $$P_S = P_A^o X_A + P_B^o X_B$$
$$= 1350\left(\frac{2}{3}\right) + 300\left(\frac{1}{3}\right)$$
$= 450 \times 2 + 100$
= 1000 torr
$$X_A = \frac{\frac{1000}{50}}{\frac{1000}{50} + \frac{700}{70}} = \frac{20}{20+10} = \frac{2}{3}$$
$P_S = P_{external}$
 Option (2)

69)

–ve deviation
 B.P. highest
Option (3)

70) $\frac{1}{Y_A}$ vs $\frac{1}{X_A}$

$$Y_A = \frac{P_A^o X_A}{(P_A^o - P_B^o)X_A + P_B^o}$$

$$\frac{1}{Y_A} = \frac{P_A^o - P_B^o}{P_A^o} + \frac{P_B^o}{P_A^o}.\frac{1}{X_A}$$

Slope $= \frac{P_B^o}{P_A^o}$

71)

A : Correct
R : correct and explain

72) $A = \frac{w}{80}, \; B = \frac{w}{120}$

$$X_A = \frac{\frac{w}{80}}{\frac{w}{80} + \frac{w}{120}}$$

$$= \frac{\frac{1}{80}}{\frac{200}{80\times120}} = \frac{120}{200} = \frac{3}{5}$$

$$X_B = \frac{2}{5}$$

$$P_S = 40\left(\frac{3}{5}\right) + 30\left(\frac{2}{5}\right)$$

$= 24 + 12 = 36$

$$\frac{P_S}{6} = 6$$

73) $P_{O_2} = Kn_{O_2}.X_{O_2}$

$0.2 \times 1 \times 10^5 = 2.5 \times 10^9 \; . \; X_{O_2}$

$$X_{O_2} = \frac{2 \times 10^4}{2.5 \times 10^9} = \frac{20}{2.5} \times 10^{-6}$$

$$= \frac{200}{25} \times 10^{-6} \quad = 8 \times 10^{-6}$$

$$X_{N_2} = \frac{P_{N_2}}{Kn_{N_2}} = \frac{0.8 \times 1 \times 10^5}{6 \times 10^9}$$

$$= \frac{8}{6} \times 10^{-5}$$

$= 1.33\times 10^{-5}$

$\frac{X_{N_2}}{X_{O_2}} = \frac{n_{N_2}}{n_{O_2}} = \frac{1.33 \times 10^{-5}}{8 \times 10^{-6}}$ in dried sample

$$\frac{n_{N_2}}{n_{O_2}} + 1 = \frac{1.33 \times 10^{-5}}{8 \times 10^{-6}} + 1$$

$$\frac{n_{N_2} + n_{O_2}}{n_{O_2}} = \frac{(13.3 + 8) \times 10^{-6}}{8 \times 10^{-6}}$$

$$\frac{n_{O_2}}{n_{N_2} + n_{O_2}} = \frac{8}{21.3}$$

$$\frac{n_{O_2}}{n_{N_2}+n_{O_2}} \times 100 = \frac{8}{21.3} \times 100$$
= 37.5%

74)

P→2
Q→ –ve deviation → 1
R → +ve deviation → 3
S → –ve deviation→ 1

75)

(1) i = 5 ; i = 2
(2) i = 5 ; i = 5
(3) i = 2 ; i = 3
(4) i = 5 ; i = 5

76)

	A →	2B +	3C
t = 0 conc.	20	0	0
t = t	5	30	45

$$K = \frac{[A_0]-[A_t]}{t}$$
$$10 = \frac{20-5}{t}$$
$$t = \frac{15}{10} = 1.5 \text{ min}$$
= 1.5 × 60 = 90 sec

77)

$T_{75\%} = 2 \quad T_{50\%} \quad A = 2$
$T_{87.5\%} = 3 \quad T_{50\%} \quad B = 3$
$T_{99\%} = 2 \quad t_{90\%} \quad C = 2$
A+ B + C = 7

78)

It is straight line so zero order

79) $$\frac{E_a}{T_1} = \frac{E_{aw}}{T_2}$$
$$\frac{E_a}{600} = \frac{E_a - 20}{500}$$
$5E_a = 6E_a - 120$

E_a = 120 kJ/mol

80)

$$\log K = \log A - \frac{E_a}{2.303RT}$$

$$\frac{E_a}{2.303R} = 1$$

$E_a = 2.303 \times 2 = 4.606$

81)

$$\log 100 = 14 - \frac{1.2 \times 10^4}{T}$$

$$2 = 14 - \frac{1.2 \times 10^4}{T}$$

$$\frac{1.2 \times 10^4}{T} = 12$$

$T = 1 \times 10^3 = 1000$ K

82) Order = $1 + \frac{2}{3} = \frac{5}{3}$

Unit of k = $(\text{mol lit}^{-1})^{1-n}\ \text{sec}^{-1}$

$= (\text{mol lit}^{-1})^{1-5/2}\ \text{sec}^{-1}$

$= (\text{mol lit}^{-1})^{-3/2}\ \text{sec}^{-1}$

83) $\frac{t}{t_{1/2}}$

$$\frac{t_{15/16}}{t_{1/2}} = \frac{\frac{2.303}{k}\log\frac{16}{1}}{\frac{2.303}{k}\log\frac{2}{1}} = \frac{4\log 2}{\log 2} = 4$$

84) $k_1 = Ae^{-\frac{E_a}{RT}}$ ...(1)

$k_{wc} = Ae^{-\frac{E_{awc}}{RT}}$ ...(2)

$$\frac{k_{wc}}{k_1} = e^{\frac{E_a - E_{awc}}{RT}} = e^{\frac{E_a - \frac{3}{4}E_a}{RT}} = e^{\frac{1}{4}\frac{E_a}{RT}}$$

$= e^{\frac{1}{4}\times\frac{10\times 1000}{\frac{25}{3}\times 300}}$

$= e^3$

85)

(1) $\alpha = \frac{A_O - A_t}{A_O}$

$$= \frac{A_O(1-e^{-kt})}{A_O} = 1 - e^{-kt}$$

(2) Unit of A is time^{-1}

(3) wrong it is not stright line
 Option (4)

86)

Concentration of product increase exponentially and concentration of reactant decreases exponentially but $t_{1/2}$ is constant.
So (1) (3) (4) all correct

87)

$$\log K = \log A - \frac{Ea}{2.303\,RT}$$

$$\frac{Ea}{2.303\,R} = \tan\left(\tan^{-1}\frac{1}{2.303}\right)$$

$$\frac{Ea}{2.303\,R} = \frac{1}{2.303}$$

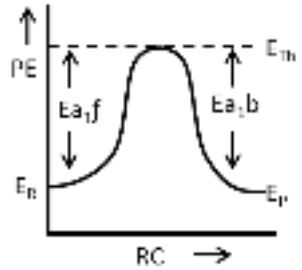
$E_a = R = 2$ cal/mol

88)

Explain Question:
To identify that the given statement is correct or not.

Concept:
Collision theory

Solution:
⇒ collision between those molecules having sufficient amount of energy to cross energy barrier and having proper orientation lead to product formation. so statement -I is correct.
⇒ According PE vs R.C. diagram



$E_{Th} = E_R + Ea_{1f} = E_P + Ea_{1b}$
So statement II is also correct.

Final answer: Option (1)

89) $P_{N_2} = 5 \times 0.8 = 4$ atm

$4 = 1 \times 10^5 \times \frac{n_{N_2}}{10}$

$n_{N_2} = 4 \times 10^{-4}$ mole.

90) $\frac{P_A^o(1-x)}{P_B^o.x} = \frac{y}{1-y}$

$P_A^o(1-x) = P_A^o y - xy[P_A^o - P_B^o]$

$P_A = P_A^o y - xy[P_A^o - P_B^o]$

$P_A^o = 235$

$P_A^o - P_B^o = 125$

$P_B^o = 110$

BIOLOGY

91) NCERT(XI) ; Page No. 91

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105) NCERT (XI) : Page No. # 102

106) NCERT XI Pg.# 71

107) NCERT XI Pg.# 72

108) NCERT XI Pg.# 77

109) NCERT XI Pg.# 77

110) NCERT XI Pg.# 76

111) NCERT XI Pg.# 74

112) NCERT XI Pg.# 74-76

113) NCERT XI Pg.# 63, 64, 69

114) NCERT XI Pg.# 62-63

115) NCERT XI Pg.# 61

116) NCERT XI Pg.# 64

117) NCERT XI Pg.# 69

118) NCERT XI Pg.# 59

119) NCERT XI Pg.# 62

120) NCERT XI Pg.# 65

121) NCERT-XI Pg#108

122) NCERT-XI Pg#109,110

123)

NCERT-XI Pg#106,107
Palmitic acid is an example of= Saturated fatty acid.
It is a type of fatty acid that does not contain any double bonds between carbon atoms, making it "saturated" with hydrogen atoms.
CORRECT OPTION = 3

124) NCERT-XI Pg#108,109

125)

NCERT-XI Pg#106,110
Explanation:

A. **Statement (1):** "Palmitic acid has 20 carbon atoms excluding carboxyl carbon." **False.** Palmitic acid ($C_{16}H_{32}O_2$) has 16 carbon atoms including the carboxyl carbon, meaning it has 15 carbon atoms excluding the carboxyl carbon.
B. **Statement (2):** "In solutions of different pHs, the structure of amino acid changes." **True.** Amino acids can change their ionization state depending on the pH of the solution, which affects their structure and charge. This is why they can exist in different forms (zwitterions, cations, or anions) at different pH levels.
C. **Statement (3):** "In a polysaccharide chain, left end is called reducing end and right end is called non-reducing end." **False.** In a polysaccharide chain, the reducing end is the end with a free anomeric carbon (usually at the right end), and the non-reducing end is (usually at the left end).
D. **Statement (4):** "Cellulose forms helical secondary structure." **False.** Cellulose has a linear, unbranched structure, not a helical structure like starch.

The Correct Answer is option 2

126) NCERT-XI Pg#110,111

127) NCERT-XI Pg#109,111

128) NCERT-XI Pg#104,108

129) NCERT-XI Pg#117

130) NCERT-XI Pg#117,118

131)

NCERT-XI Pg#116

132)

NCERT XI Pg.# 118

133) NCERT XI Pg. # 112-118

134)

NCERT-XI, Pg.# 116

135) NCERT-XI Pg#115,116

136) NCERT, Pg # 134

137) NCERT-XI Pg#143,146,147

138) NCERT-XI Pg#141

139)

NCERT-XI, Pg. # 150

140) NCERT-XI, Pg. # 134, 136, 137, 140

141) NCERT-XI Pg#140

142) NCERT-XI Pg#135,137

143) NCERT-XI Pg.#139,140,141,142,145

144) NCERT-XI, Pg # 137

145)

NCERT-XI Pg # 144
For 1 CO_2 reduction step = 2ATP,
regeneration step = 1 ATP
Difference is 1 ATP

146)

NCERT-XI Pg#144

147) NCERT-XI Pg#143,146,147
The Enzyme that help in Primary carboxylation in C_3 plants and secondary carboxylation in C_4 plants is RUBISCO.

148) NCERT-XI Pg#138,139,140

149) NCERT-XI Pg # 145, 146

150) NCERT-XI Pg#146

151) NCERT-XI Pg#159

152) NCERT-XI Pg#156

153)

NCERT XI, Pg # 160

154)

NCERT-XI Pg#156,157,160

155)

NCERT-XI Pg#159
Explanation: In the Krebs cycle (TCA cycle), the reduction of cofactors occurs as follows:

NAD^+ is reduced to NADH + H^+ at three steps:

A. Isocitrate → α-ketoglutarate
B. α-ketoglutarate → Succinyl-CoA
C. Malate → Oxaloacetate

B. FAD is reduced to $FADH_2$ at one step: Succinate → Fumarate

Correct Answer: 2. Three and one

156) NCERT-XI Pg#156

157) NCERT-XI Pg#154,162

158)

NCERT-XI Pg. # 161, 162

159)

NCERT-XI Pg#156,157

160) NCERT-XI Pg.#155,158,159,160

161) NCERT-XI, Pg # 158,159

162) NCERT XI, Pg. # 160

163) NCERT-XI, Pg. # 156

164) NCERT-XI Pg. # 156,164,165

165) It happened due to availability of oxygen and switching to aerobic respiration.

166) NCERT-XI Pg # 177

167)

NCERT-XI Pg#175
The correct answer is:
2. Auxin
Explanation:

A. Auxins are the hormones primarily responsible for promoting root formation from the cut ends of stems. When applied to the cut surface of a stem, auxins stimulate the production of roots, aiding in the propagation of plants through cuttings.

B. Common types of auxins used for this purpose include Indole-3-acetic acid (IAA) and Indole-3-butyric acid (IBA).

168)

NCERT-XI Pg#177
SOLUTION
Sex Expression in Some Plants: In some cucurbits (like cucumbers), ethylene can influence the formation of male and female flowers. Typically, high levels of ethylene promote male flower production, whereas low levels may promote female flowers.

169)

NCERT-XI Pg#172
Explain Question: The question is asking for the term used when mature living cells regain their ability to divide.

Concept: Dedifferentiation

Solution:

The living differentiated cells, that by now have lost the capacity to divide can regain the capacity of division under certain conditions. This phenomenon is termed as dedifferentiation.
For example, formation of meristems – interfascicular cambium and cork cambium from fully differentiated parenchyma cells.

Final answer: Option 1

170) NCERT-XI Pg. # 176

171)

NCERT-XI Pg#167
Explain Question: The question asks to analyse the given statements and select the correct options.

Concept: Growth.

Solution:

Assertion (A) : Plants retain the capacity for unlimited growth throughout their life.

Reason (R) : Generally growth is accompanied by metabolic processes that occur at the expense of energy.

Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

So, the correct answer is: 1.

Final answer: Option 1

172) NCERT-XI, Pg. # 171

173) NCERT-XI Pg#166,171

174)

NCERT-XI Pg#175,177,178
Explain Question: The question is asking to choose the option that correctly categorises plant growth regulators (Auxin, Abscisic acid, Ethylene) according to their physiological effects.

Concept: Physiological Effects of Plant Growth Regulators

Solution:

Auxin → induces parthenocarpy (seedless fruits)
Abscisic acid (ABA) → involved in seed maturation and acts as a stress hormone
Ethylene → used for thinning of cotton (removal of excess flowers/fruits), promotes ripening and abscission

Only Option 1 correctly matches these effects.

Final answer: Option 1

175) NCERT-XI Pg # 172, 173

176)

NCERT-XI Pg#178

177) NCERT-XI, Pg. # 173

178) NCERT-XI, Pg. # 176

179) NCERT-XI Pg. # 172, 178

180) NCERT-XI Pg.#174, 175, 176, 177